# DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT SYSTEM

A project report submitted in partial fulfilment of the requirement for the award of the degree for

## BACHELOR IN INFORMATION TECHNOLOGY

by
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# PREFACE

The Development of University Event Management System was developed as part of our final year project to provide a structured and efficient solution for organizing and managing university events.

The system aims to centralize event creation and announcements, registration, and administrative oversight within a unified digital platform, improving coordination between students, organizers, and university management.

Traditional event management methods in many institutions rely on manual processes and scattered communication channels, often resulting in delays, inconsistencies, and reduced student engagement. This project addresses these challenges by offering a streamlined online environment that enhances transparency, accessibility, and overall event organization.

The development of this system allowed us to apply theoretical knowledge to a real-world problem while strengthening our skills in system design and database development, web implementation, and teamwork. We hope that this platform will contribute to a more organized and engaging campus experience for all university stakeholders.

## ACKNOWLEDGEMENT

We would like to express our sincere appreciation to everyone who contributed to the successful completion of this project, Development of `University Event Management System. We are deeply grateful to our project supervisor Dr. Jayabrabu Ramakrishnam for his continuous guidance, constructive feedback, and valuable support throughout all stages of the work. His expertise and Encouraging played a crucial role in shaping the direction and quality of this project.

We also extend our thanks to the faculty and staff of the College of Engineering and Computer Science for providing resources and academic environment necessary to carry out this work. Their commitment to supporting student projects has greatly enriched our learning experience.

Finally, we acknowledge the collective efforts of our team members, whose collaboration, dedication, and shared commitment made this project possible. Each member contributed meaningfully to the analysis, design, and development phases, ensuring the project meets its object

# Table of Content

# List of Figures



# List of Tables

# ABSTRACT

The Development of University Event Management System is designed to provide an integrated and efficient solution for organizing, coordinating, and monitoring university events within a centralized digital platform. Many academic institutions still rely on manual procedures and dispersed communication channels such as printed announcements, social media posts, or individual emails which often lead to scheduling conflicts, low participation, and administrative inefficiencies. This project addresses these limitations by offering a structured web-based system that enhances accessibility, transparency, and coordination among students, event organizers, and ad- ministrative departments. The platform enables organizers to create and manage event proposals, publish announcements, and track attendance, while allowing students to easily browse, register for, and engage with events relevant to their academic and extracurricular interests. Administrators benefit from streamlined approval workflows, improved oversight, and consolidated data that support informed decision-making. This report presents the system's objectives, requirements, feasibility assessment, and design methodology, demonstrating how a unified digital solution can significantly improve the quality and effectiveness of event management in academic environments.

Keywords:

- University Event Management
- Web-Based System
- Digital Coordination
- Event Registration
- Administrative Workflow
- Student Engagement
- Information Systems
- Event Scheduling
- System Design
- Campus Activities

# 1. INTRODUCTION

## Project Overview Statement

This project aims to design and develop a comprehensive electronic event booking system for Jazan University. The system is conceived to address the current challenges in campus event management, which often involve fragmented information, inefficient booking processes, and a lack of a unified platform for coordination. By creating a single, integrated digital platform, this project will connect three key user groups: the students who seek to discover and participate in events, the organizers (such as student clubs, departments, or faculties) responsible for planning and executing these events, and the university administration that oversees and approves them. For students, the platform will serve as a centralized hub to browse, search, and register for a wide array of university activities, from academic seminars and workshops to cultural and social gatherings, thereby enhancing their campus life engagement. For organizers, the system will provide a suite of tools to create event listings, manage promotions, handle registrations, track attendance, and communicate with attendees, significantly streamlining their workflow and reducing administrative overhead. For the university administration, the platform will offer a clear overview of all scheduled events, facilitate the approval process, enable efficient resource allocation (such as venues and equipment), and provide valuable data analytics on event participation and impact, supporting data driven decision-making. In essence, this project seeks to digitize and modernize the event management lifecycle at Jazan University, fostering a more connected, efficient, and vibrant campus community by simplifying how events are discovered, managed, and experienced Project Goals and Objectives.

## Project Goals and Objectives

### Project Goals:

1. To Centralize Campus Event Management: Create a single, authoritative source for all university-related events, eliminating information fragmentation across different departments, clubs, and communication channels.
2. To Enhance Student Engagement: Foster a more vibrant and connected campus community by making it easier for students to discover, participate in, and engage with activities that align with their interests.
3. To Streamline Administrative Processes: Reduce the manual effort and administrative overhead for event organizers and the university administration by digitizing and automating key workflows in the event lifecycle.
4. To Improve Communication and Coordination: Facilitate seamless communication between students, event organizers, and the administration, ensuring all stakeholders are informed and aligned.

### Project Objectives:

1. To design and develop a fully functional, responsive website that serves as the event booking platform, completing all core modules (Student, Organizer, and Administrator).
2. To implement a user-friendly event discovery and registration process, enabling students to find and register for an event.
3. To provide organizers with a digital toolkit that reduces the time spent on administrative tasks.

## Project Scope

1. The scope of this project is focused on the development of a web-based event booking system for Jazan University.
2. This system will provide a public interface for students to browse, search, and register for events, a secure portal for event organizers to create, manage, and track attendance for their activities, and a dedicated administrative dash- board for oversight, user management, and event approval.

3. The project scope includes direct integration with student university data to verify user identity and ensure that all services are exclusively available to enrolled university students.
4. The system focuses on managing internal, university events and provides a feature for verifying a student's reservation, either through their personal dashboard or the list of active bookings

## Limitations and Constraints

- **Constraints**

1. A Resource Constraint: The project will be developed with minimal to no financial budget. This restricts the use of premium software tools, paid cloud hosting services, or the procurement of digital assets. All developments will rely on open-source technologies and free-tier service plans Assumptions.
2. Data Privacy Constraint: The project is bound by ethical considerations regarding user data. It cannot use or store real, sensitive student or university data. All development and testing will be conducted using fictitious or mock data to ensure compliance with privacy standards. The system does not handle grading or academic evaluations, just group formation and supervision requests.

- **Limitations**

1. Scalability Limitation: The system will be tested by a limited number of users within a controlled environment. Therefore, its performance and stability under heavy load (e.g., thousands of concurrent users during a major event registration) will not be fully validated.
2. Security Limitation: While standard security practices (e.g., password hashing, input validation) will be implemented, the system will not undergo a professional, third-party security audit. It may not be robust enough to defend against highly sophisticated cyber-attacks compared to a large-scale enterprise application.
3. User Feedback Limitation: The evaluation of the system's usability and effectiveness will be based on feedback from a small, targeted group of test users. This may not capture the full spectrum of needs and preferences of the entire diverse student and faculty population of the university.

## Assumptions

- **Technical Assumptions**

1. Internet Access: All target users (students, organizers, and administrators) have reliable access to the internet to use the web-based system.
2. Device and Browser Compatibility: Users will access the system using modern, up-to-date web browsers (e.g., Chrome, Firefox, Safari) on either a desktop or mobile device. The system will not be designed to support outdated browsers
3. Hosting Environment: The project team will be able to secure a suitable hosting environment (e.g., a university server or a free-tier cloud platform) to deploy and demonstrate the final system.

- **User and Stakeholder Assumptions**

1. Digital Literacy: Students and event organizers possess a basic level of digital literacy required to navigate and use a web-based application without extensive training.
2. Willingness to Adopt: Event organizers (student clubs, departments) will be willing to use the new system and will provide accurate and timely information for their events.

# 2. REVIOUS WORKS/ LITERATURE REVIEW/CASE STUDY/BACKGROUND

## 2.1. Event Management Department at King Saud University

The Event Management Department at King Saud University Medical City (KSUMC) plays a pivotal role in the planning, coordination, and execution of a wide array of events within the institution. This department utilizes comprehensive project management principles to effectively organize and manage both medical and non-medical events, ranging from national and international conferences to workshops, graduation ceremonies, re- search days, and more. Its primary mission is to ensure that all events are executed smoothly, efficiently, and in alignment with the strategic goals and objectives of KSUMC.

### 2.1.1. Limitations of the Event Management Department (EMD) at KSUMC

1. Scope Restricted to Defined Event Categories The department's activities are primarily limited to its stated event types (international/national conferences, symposia, workshops, graduation events, and research days). This structured scope may prevent the department from exploring or implementing unconventional or emerging event models outside its predefined frame- work.
2. Resource and Capacity Constraints Managing a wide variety of large-scale medical and academic events requires ex- tensive coordination, staff expertise, and logistical support. The EMD may face limitations in manpower, time, or physical facilities—especially during peak aca- demic or clinical seasons.
3. Reliance on Cross-Department Collaboration Effective event execution depends heavily on cooperation with clinical departments, administrative units, and external partners. Any delays, miscommunication, or administrative bottlenecks in other departments may hinder EMD performance.

## 2.2. Transform events management with TAM's Jovial

The event management sector in Saudi Arabia is undergoing significant growth, driven by Vision 2030 initiatives in sports, music, and culture, and is expected to surpass $1 billion by 2030 from its current valuation of about $600 million annually. This growth is being facilitated by smart technologies like TAM's digital product, Jovial, an Event Management System (EMS) designed to elevate offerings and maximize revenue. Jovial, part of TAM's Verse portfolio, is crucial for the operational success of events, enabling customers to seamlessly book and pay for various events (including single-day, multi-day, group, and festival experiences), while allowing organizers to manage their full event cat- alogia, seating maps, and secure operations effectively. Furthermore, the system, which can handle bookings for up to 250,000 attendees, provides personalized onboarding and in-built functionality for managing different membership and subscription options (yearly, bi-yearly, and monthly).

### 2.2.1. Limitations of The Transform events management with TAM's Jovial

1. Learning Curve: As a comprehensive, end-to-end system, Jovial likely has a steep learning curve for new users or staff members. Utilizing its full range of features—from seat mapping to membership management—requires thorough training and time to master.
2. Dependency on a Single Vendor: Relying on one comprehensive platform for all end-to-end processes creates a vendor lock-in, meaning the organizer's operation is highly dependent on TAM for system updates, support, and feature development.
3. Integration Challenges: While the source text mentions integration with financial and reporting systems, there could be limitations in integrating with every possible external tool (e.g., specific marketing platforms, unique third-party vendor systems) that an organizer might already use.

## 2.3. LITERATURE REVIEW

**Event Booking & Reservation Systems**

Previous research highlights the rapid shift toward web-based platforms for managing events, bookings, and registrations. These digital systems allow users to browse event listings, register online, and receive instant confirmations, significantly improving accessibility and user convenience. For organizers, they provide essential tools such as event creation interfaces,[6] participant tracking dashboards, real-time schedule updates, and centralized data management. For example, a study titled "Online Events Booking and Reservation System" found that digital reservation platforms greatly reduce administrative effort by enabling users to make bookings remotely at any time, eliminating delays associated with manual processing. This research also demonstrated that centralized booking records minimize errors that commonly arise from paper-based or email-based systems [7]. Similarly, a web-based venue booking system evaluated by the International Journal of Research Publication and Reviews showed that automated availability checking, approval workflows, and confirmation generation significantly decreased double bookings and im-proved coordination across departments (IJRPR, 2023). Additional studies on campus event management platforms have emphasized that integrating event announcements, registration, and attendance tracking into a single portal enhances student engagement and reduces operational workload. These systems also provide valuable analytics for decision-making, helping universities understand participation trends and resource requirements (IJNRD, 2025). Compared to traditional methods—such as spreadsheets, handwritten registration forms, or scattered email announcements, these digital solutions centralize information, reduce human error, increase efficiency, and offer better monitoring of events. This body of prior work forms a strong foundation for the proposed UNIVERSITY EVENT MANAGEMENT SYSTEM, which extends these capabilities by incorporating administrative approval workflows, support for diverse academic and non-ticketed events, and advanced tracking features tailored to the needs of Jazan University.

# 3. EASIBILITY STUDY

## 3.1.Purpose of The Feasibility Study

The purpose of the feasibility study for the DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT SYSTEM is to conduct a thorough and objective analysis to determine if the proposed project is viable, practical, and beneficial before committing significant resources to its development. This study acts as a critical checkpoint, providing a data-driven foundation for the decision to proceed, modify, or reconsider the project.

The primary goals of this feasibility study are to:

• Assess Technical Viability: To determine if the project can be successfully built and implemented using the available technical resources. This includes evaluating the project team's skills in the chosen technologies (e.g., web development frameworks, databases) and confirming that the necessary software and hardware are accessible within the project's constraints.

• Analyze Operational Practicality: To determine if the proposed system will fit into the existing workflows of the university and be accepted by its intended users (students, organizers, and administrators). This involves assessing whether the system solves a real problem and if users will be willing and able to adopt it effectively.

Ultimately, the purpose of this study is to provide a clear recommendation to the project stakeholders. By systematically evaluating these key areas, the feasibility study ensures that the DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT SYSTEM is founded on a realistic and well justified plan, thereby maximizing its potential for successful completion and positive impact.

## 3.2.Justification for The Proposed System

The justification for developing the DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT is rooted in the need to address critical inefficiencies in the current campus event management process and to provide significant, measurable benefits to the entire university community. The existing methods, which rely on a disjointed combination of manual paperwork, scattered email announcements, and various social media posts, are no longer adequate for a vibrant and growing institution like Jazan University.

The proposed system is justified on the following grounds:

• Resolving Inefficiency and Reducing Administrative Burden: Currently, event organizers and administrative staff spend excessive time on manual tasks such as processing paper forms, managing registration lists in spreadsheets, and answering repetitive inquiries from students. The DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT will automate these core processes—event submission, approval, registration, and communication—freeing up valuable time for staff and organizers to focus on more strategic, value-added activities.

• Enhancing the Student Experience and Engagement: A key challenge for students is the lack of a single, reliable source of information about campus activities. Important events are often missed simply because students are unaware of them. This system will create a centralized, easily searchable hub for all university events, empowering students to discover opportunities that align with their academic and personal interests. This increased visibility and ease of access will directly lead to higher participation rates and a more connected campus life.

• Improving Communication and Coordination: The current fragmented communication channels lead to misinformation and scheduling conflicts. For instance, two clubs might unknowingly plan events for the same time and venue. The DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT will serve as a single source of truth, facilitating clear communication between organizers, students, and the administration. The built-in approval workflow for events and resource booking will prevent conflicts and ensure all activities are properly coordinated.

The DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT is not merely a technological upgrade; it is a strategic solution to a pressing operational problem. It is justified by its potential to create a more efficient, transparent, and engaging campus environment, aligning directly with the university's mission to foster a dynamic and supportive academic community The proposed system is justified on the following grounds:

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## 3.3.Economic Feasibility

The economic feasibility for the DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT is very positive. This is because the project has very low financial costs, while offering many valuable benefits to the university

**Costs:**

The project will cost almost no money.

Software: All the tools needed to build the website are free and open source.

Hardware: We will use our existing personal computers.

Hosting: We can use a free hosting service to run the website.

The only real cost is the time and effort that the project team will put into designing, building, and testing the system.

**Benefits:**

The benefits of the system are significant and provide great value.

Saves Time: The system will automate many manual tasks, saving a lot of time for both student organizers and university staff.

Improves Student Life: By making it easy to find and join events, the system will help create a more active and connected campus community.

Better Decisions: The system will give the university useful data about which events are popular, helping them make smarter choices about future activities.

## 3.4. Technical Feasibility

The DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT is technically feasible. We have the necessary technology, skills, and tools to build it.

• Technology: We will use common and free web technologies. For the website's appearance, we will use HTML-5, CSS-5, and JavaScript. For the system's logic, will use php to make the connection to the database, and for storing data, we will use a reliable database like MySQL. These are all standard tools for building websites.

• Team Skills: Our project team has the required programming knowledge to develop this system. We are also capable of learning any new tools needed for the project.

• Equipment: We only need our personal computers and free software that we can download. No special or expensive hardware is required to build and run the website.

## 3.5. Desired System Functionality

The DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT SYSTEM is envisioned as a comprehensive web platform with distinct functionalities tailored to its three primary user groups: Students, Event Organizers, and Administrators. The desired functionalities are outlined below to provide a clear understanding of the system's intended capabilities.

1. **General System Functionality**

   - User Authentication: A secure login and logout system to protect user accounts and data.
   - Dashboard: A personalized homepage for each user type, providing an overview of relevant information and quick access to key features.

2. **Student Functionality**

- Event Discovery: Browse a list or calendar view of all upcoming, approved university events.
- Search and Filter: Search for events by keyword, category (e.g., Academic, Sports, Cultural), date, or organizer to find relevant activities.
- View Event Details: Access a dedicated page for each event with comprehensive information, including description, time, location, and organizer details.
- Event Registration: A simple, one-click process to register for an event.

- Manage Registrations: View a personal list of all events the student has registered for and their status.

**3. Event Organizer Functionality**

- Event Creation: A step-by-step form to submit a new event proposal with all necessary details (name, description, date, location, etc.).
- Event Management: The ability to view, edit, or cancel events that they have created.
- Registration Management: View a real-time list of students who have registered for their events.

**4. Administrator Functionality**

- User Management: The ability to create, view, and deactivate accounts for authorized event organizers.
- Event Approval: A central dashboard to review event submissions from organizers and approve or reject them before they are published to students.
- System Oversight: A comprehensive dashboard to view all system activities, including a list of all events, user statistics, and registration data.

These desired functionalities form the core of the DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT, ensuring it meets the needs of all stakeholders and effectively streamlines the event management process at the university.

# 4. PROJECT PLAN

## Work Breakdown Structure

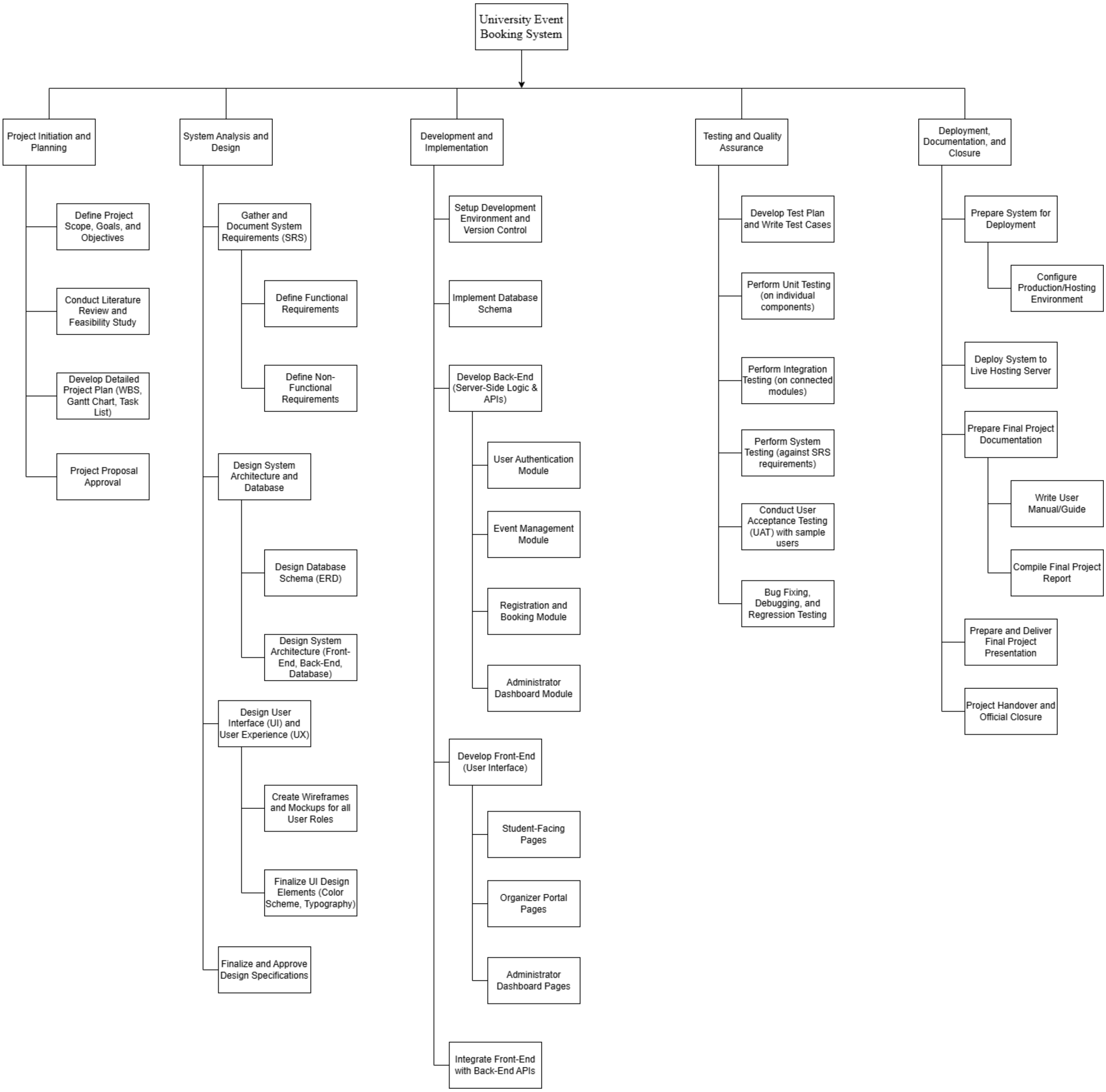


*Figure 1 Work Breakdown Structure*

## Activity and Task List

| Phase | Key Tasks | Estimated Time |
|---|---|---|
| **1. Planning** | • Finalize project goals and scope.<br>• Research other systems and write the feasibility study.<br>• Create the project plan and timeline. | 1 Week |
| **2. Design** | • Define all system features and requirements.<br>• Design the database structure.<br>• Create simple sketches (wireframes) for all website pages. | 2 Weeks |
| **3. Development** | • Set up the development tools and databases.<br>• Build the server-side logic (the "brain" of the system).<br>• Create the website pages (the "face" of the system).<br>• Connect the website pages to the server logic. | 6 Weeks |
| **4. Testing** | • Write a plan for testing all features.<br>• Test the system to find and fix any bugs.<br>• Having a small group of students and staff use the system and give feedback. | 2 Weeks |
| 5. **Deployment & Finalization** | • Put the finished system on a live web server.<br>• Write a simple user guide.<br>• Prepare the final project report and presentation. | 2 Weeks |

Table 1 Activity and Task List

# Gantt Charts

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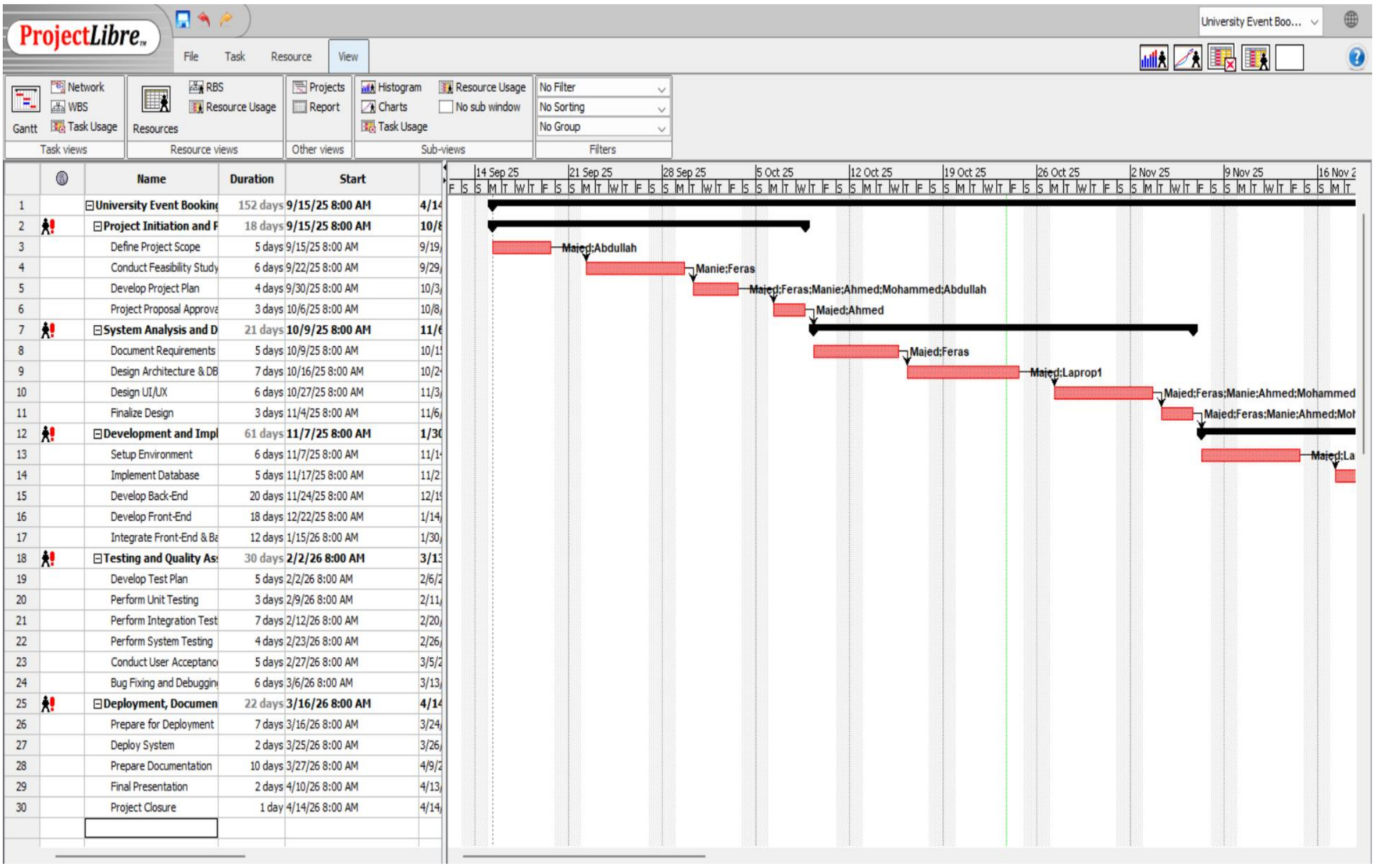



*Figure 2 Gantt Charts*

# 5. SRS – SYSTEM REQUIREMENTS SPECIFICATION

## 5.1. Development Methodology

The DEVELOPMENT OF UNIVERSITY EVENT MANAGEMENT SYSTEM will be developed using the reuse methodology. Figure 3 This iterative approach is chosen for its flexibility and suitability for academic projects where requirements may evolve.

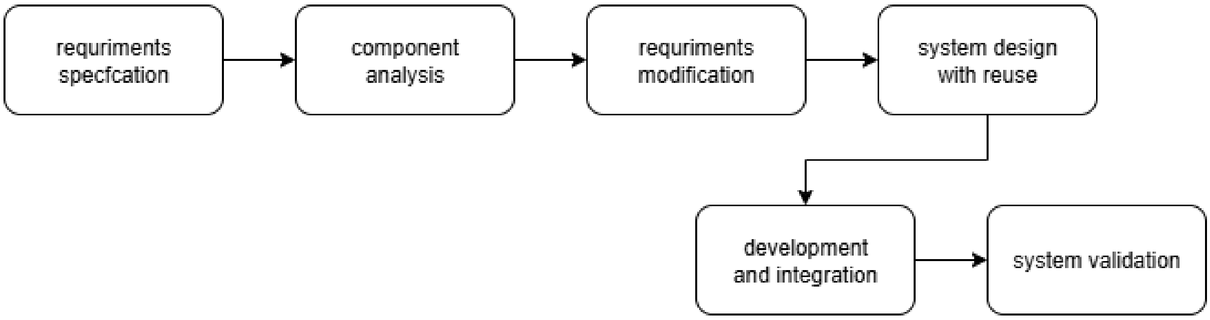



Figure 5.1: Reuse model

The project will be broken down into small, manageable development cycles called" sprints." Each sprint will typically last 1-2 weeks and will follow a cycle of planning, design, development, testing, and review.

**This methodology allows for:**

- Continuous Feedback: Regular feedback from the project supervisor and potential users can be incorporated throughout the development process.
- Flexibility: The team can adapt to changes and refine requirements as the project progresses.
- Early Delivery: A working version of the system with core features can be developed early, allowing for more thorough testing and validation.
- Risk Mitigation: Problems are identified and addressed early in smaller increments, reducing the risk of major project failures.

## 5.2. System Requirements Analysis

| Functional Requirements | Non-Functional Requirements |
|---|---|
| **User Management** | Performance |
| **Event Management** | Security |
| **Event Discovery and Registration** | Usability |
| **Administration and Oversight** | Reliability |

## 5.3. Hardware Requirements Specification

| For users | Low Requirements | Hight Requirements |
|---|---|---|
| **Processor** | Intel Core i3 | Intel Core i5 |
| **RAM** | 4 GB | 8 GB |
| **Storage** | 128 GB SSD | 512 GB SSD |
| **Network** | Wi-fi 5 | Wi-fi 7 |

## 5.4. Software Requirements Specification

| | |
|---|---|
| **Operating System** | **Windows 10/11, macOS, or a modern Linux distribution** |
| **IDE/Code Editor** | Visual Studio Code. |
| **Web Browser** | Google Chrome or Mozilla Firefox |

## 5.5. Other **Requirements Specifications**

| |
|---|
| **Data Privacy: The system will comply with basic data privacy principles.** |
| **Language: The primary language of the system will be English.** |

# 6. SYSTEM DESIGN

## Data Flow Diagram

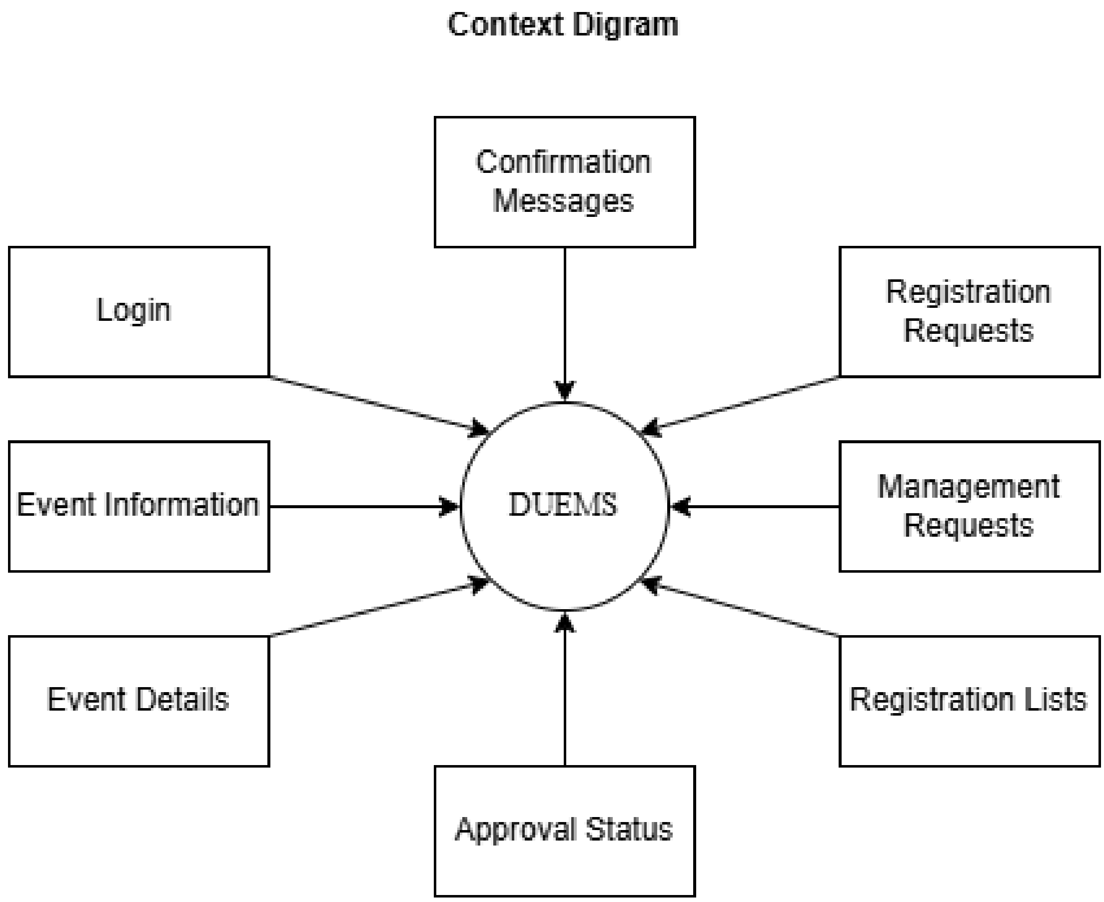



*Figure 6-1.1 Data Flow level 0*

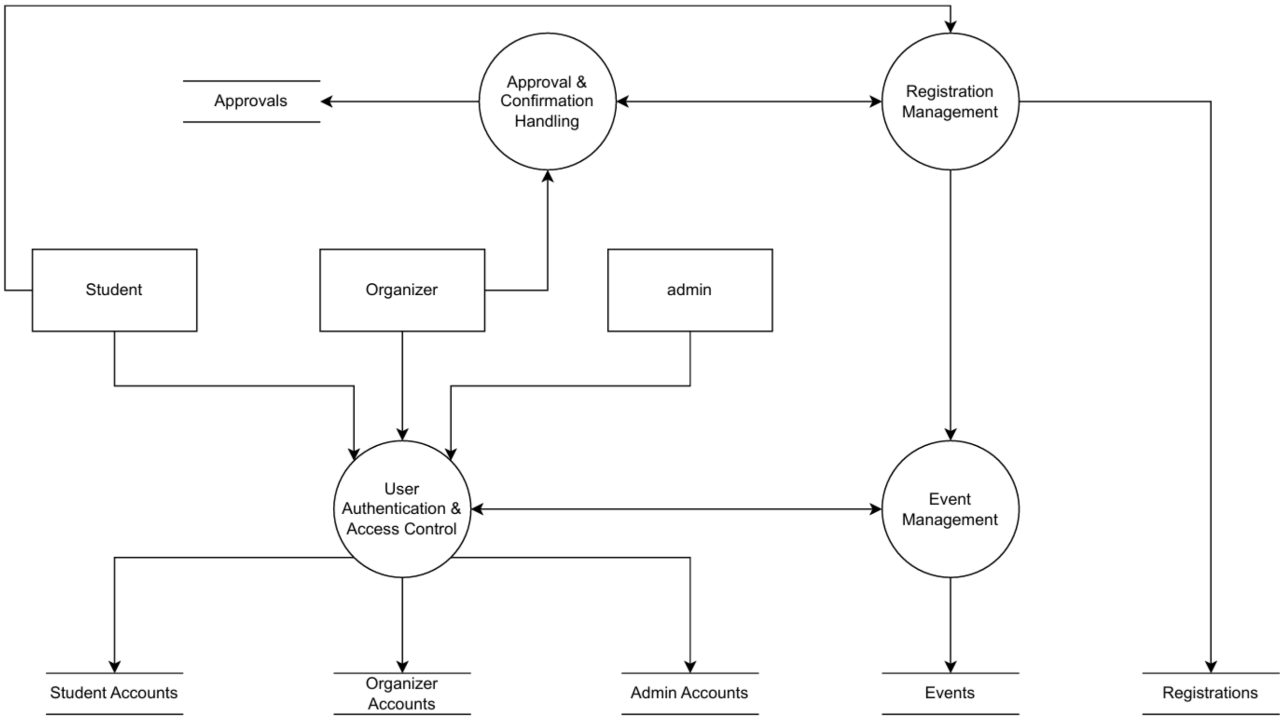


*Figure 6-1.2 Data Flow level 1*

# Class Diagram

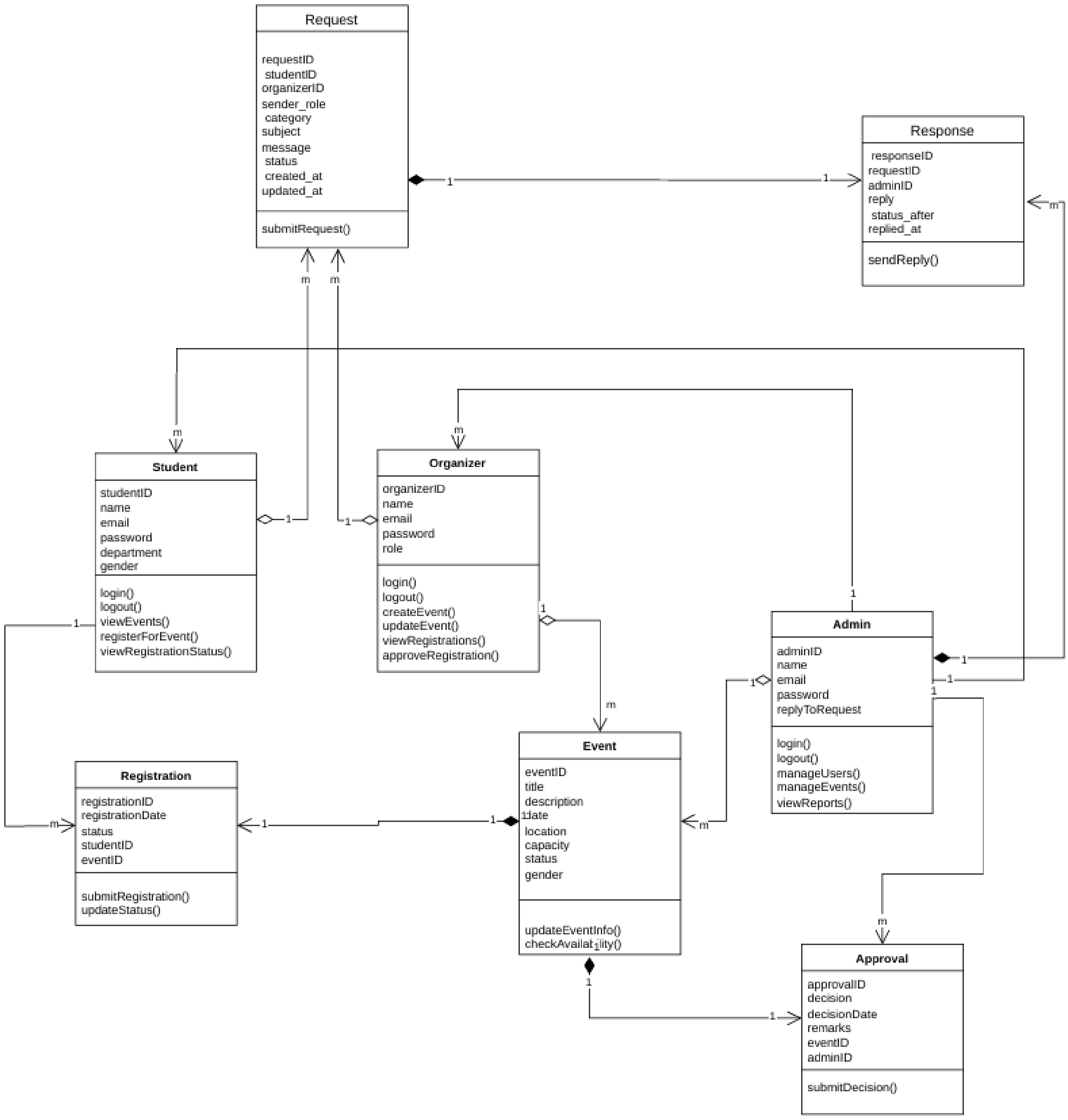



*Figure 6-2 Class Diagram*

## Use case Diagram

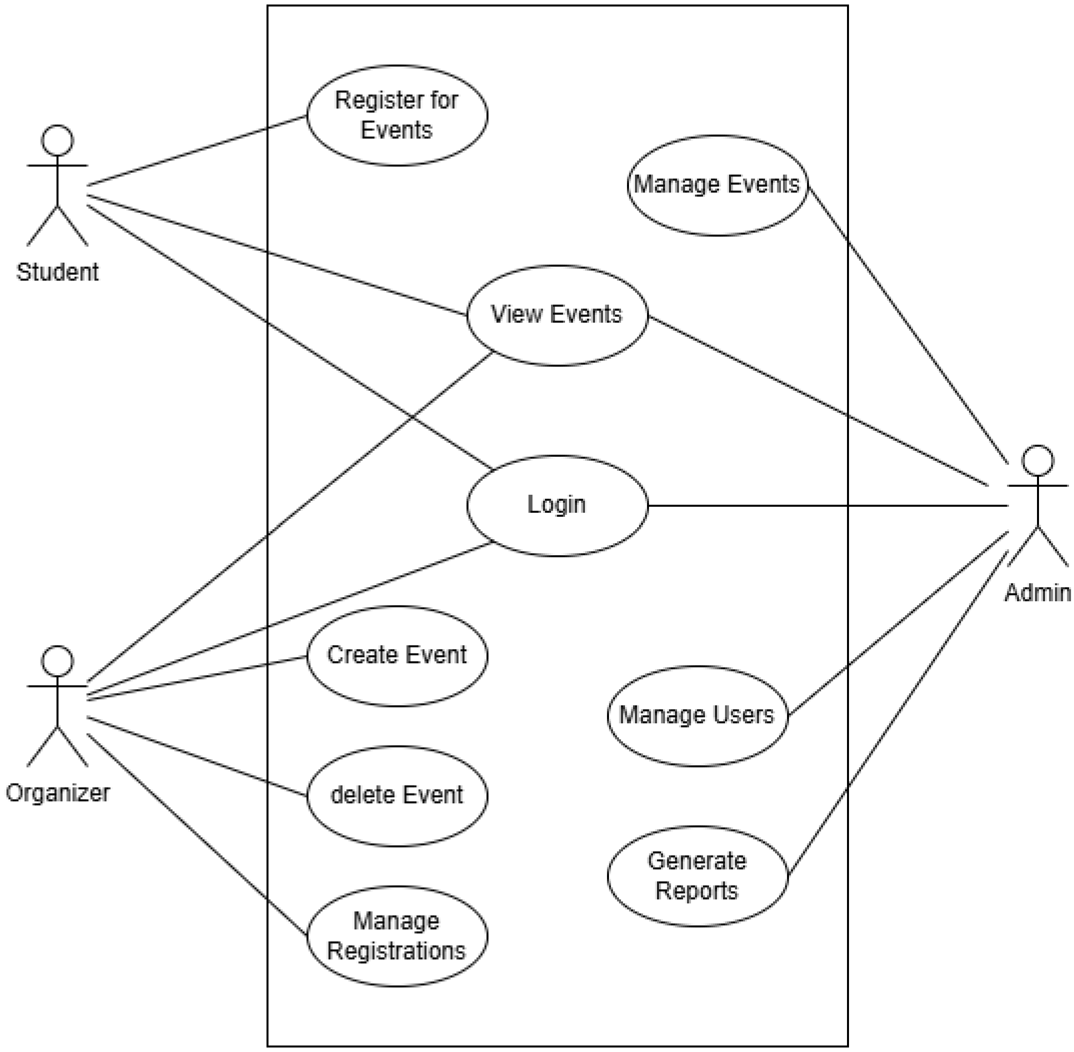



*Figure 6.3 Use case Diagram*

## Activity Diagram

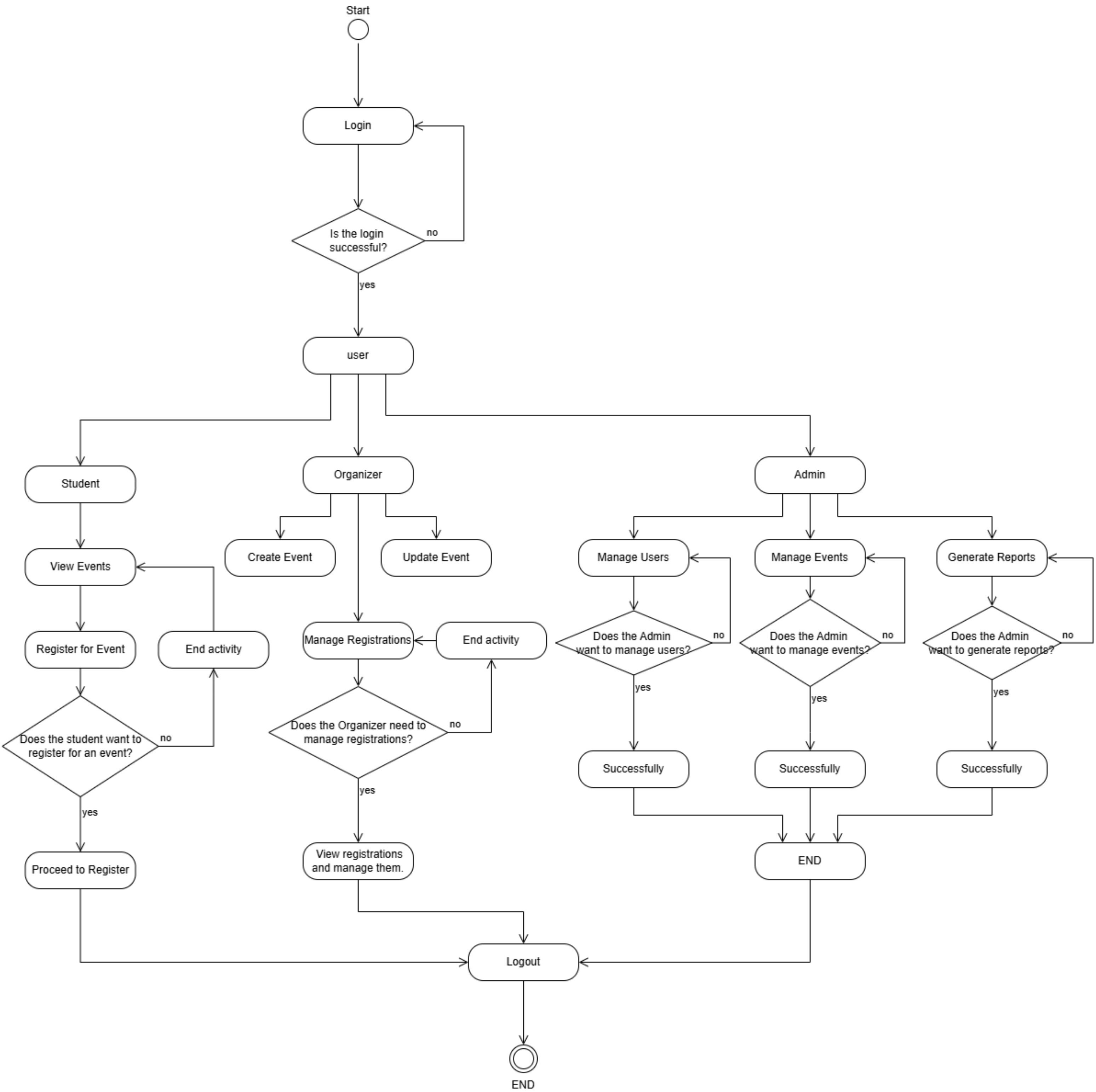



*Figure 6-4 Activity Diagram*

## Sequence Diagrams

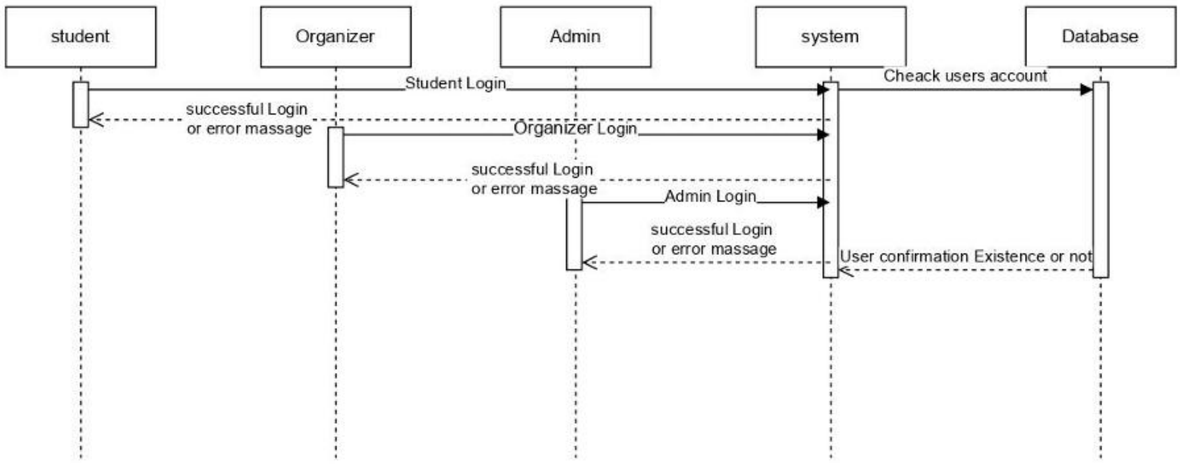



*Figure 6-5.1 Sequence Diagrams: login*

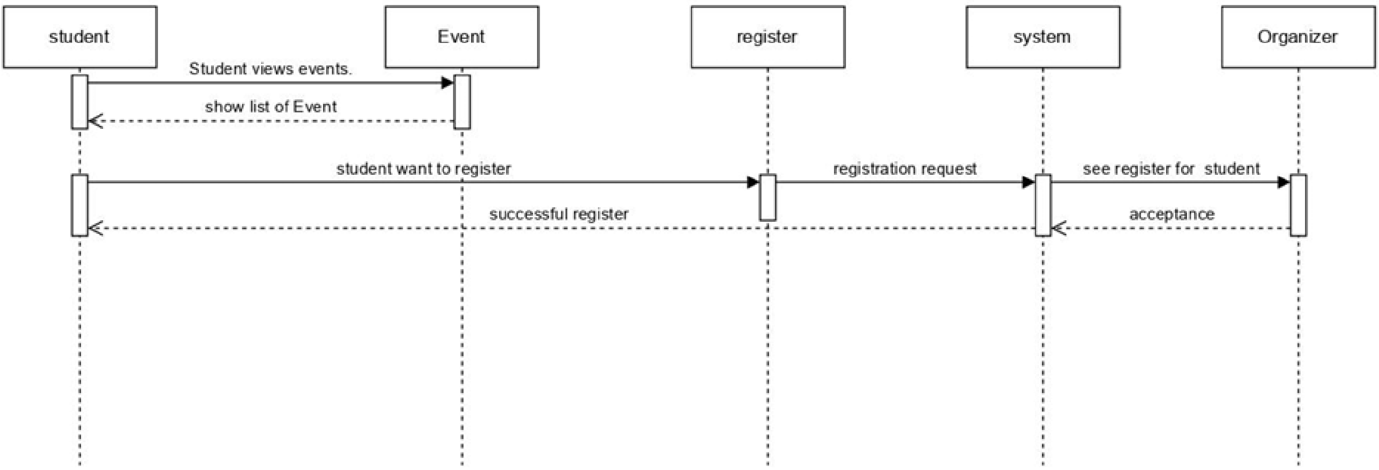



*Figure 6-5.2 Sequence Diagrams: Student event*

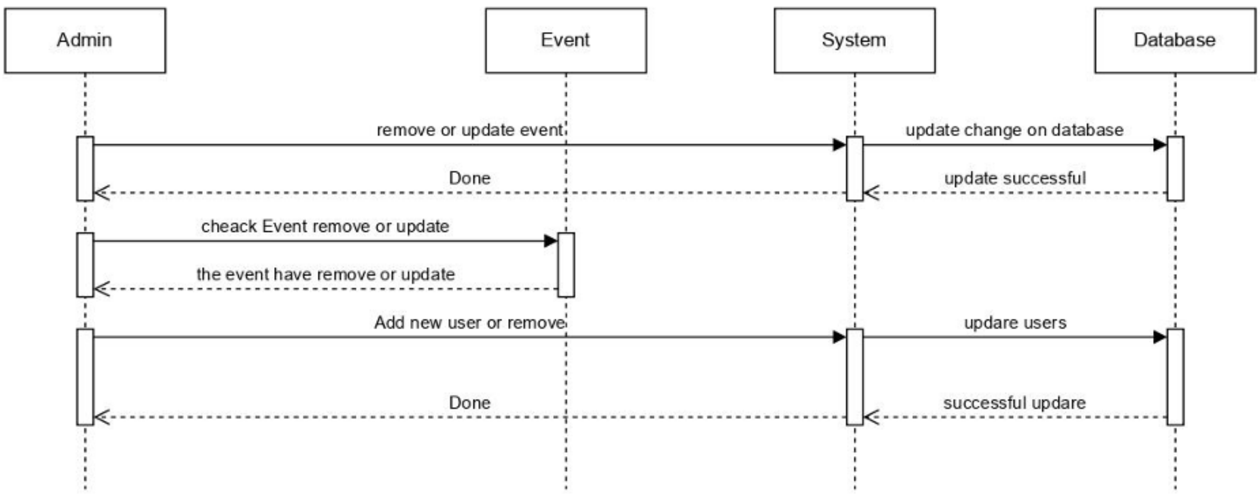



*Figure 6-5.3 Sequence Diagrams: admin-management.*

## Entity Relationship Diagram:

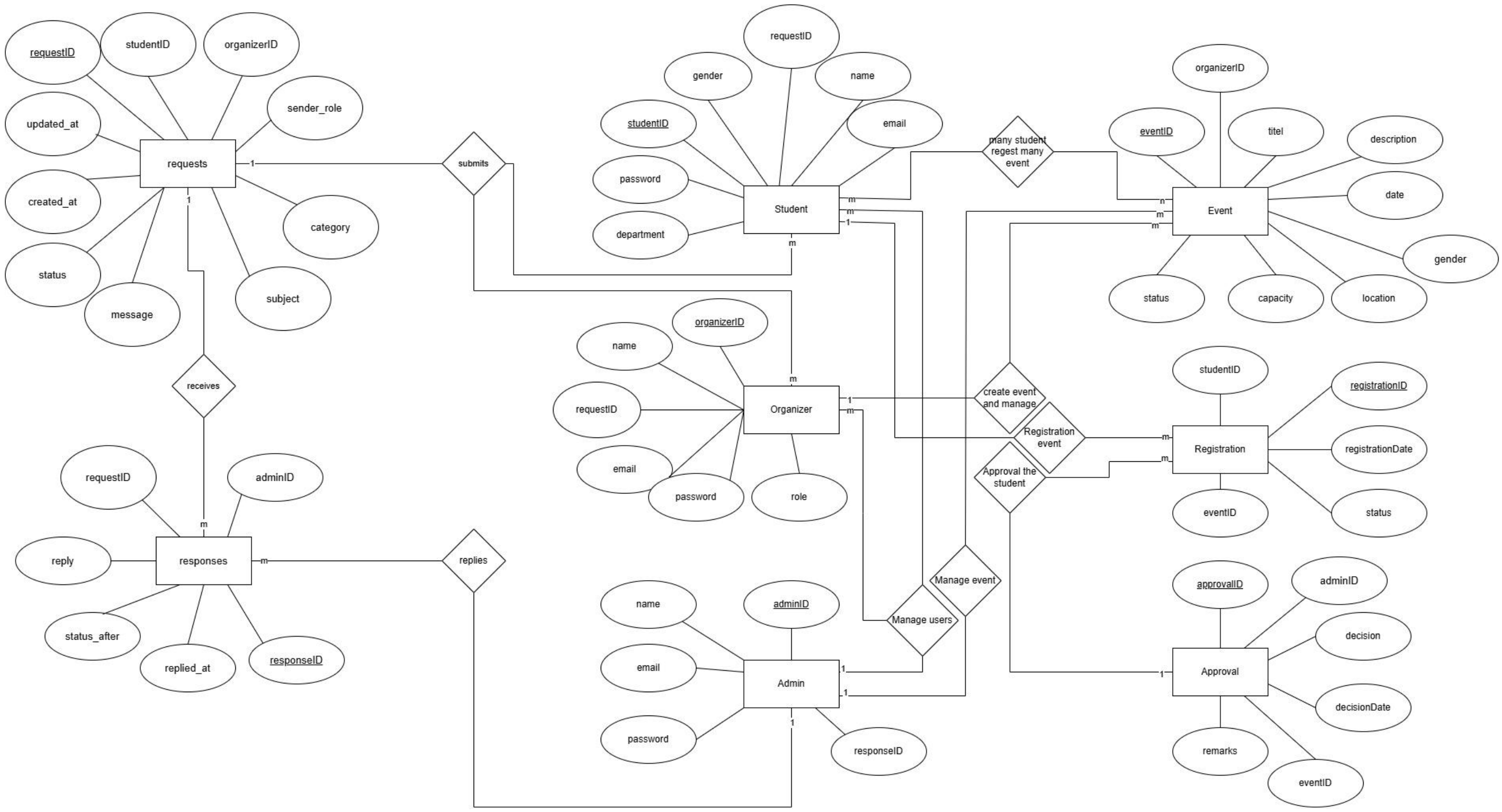



*Figure 6-6 Entity Relationship Diagram*

# Database Design:

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | adminID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 2 | name | varchar(60) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 3 | email | varchar(80) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 4 | password | varchar(255) | utf8mb4_general_ci | | No | None | | | Change Drop More |

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | organizerID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 2 | name | varchar(60) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 3 | email | varchar(80) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 4 | password | varchar(255) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 5 | role | varchar(40) | utf8mb4_general_ci | | No | None | | | Change Drop More |

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | responseID | int(11) | | | No | None | | AUTO_INCREMENT | Change Drop More |
| 2 | requestID | int(11) | | | No | None | | | Change Drop More |
| 3 | adminID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 4 | reply | text | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 5 | status_after | enum('Pending', 'Resolved') | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 6 | replied_at | timestamp | | | No | current_timestamp() | | | Change Drop More |

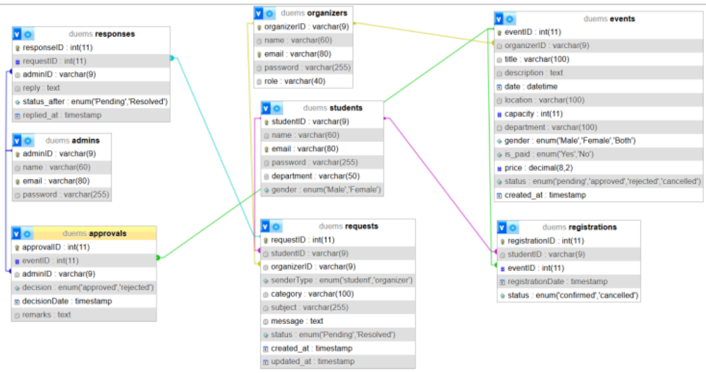
| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | requestID | int(11) | | | No | None | | AUTO_INCREMENT | Change Drop More |
| 2 | studentID | varchar(9) | utf8mb4_general_ci | | Yes | NULL | | | Change Drop More |
| 3 | organizerID | varchar(9) | utf8mb4_general_ci | | Yes | NULL | | | Change Drop More |
| 4 | senderType | enum('student', 'organizer') | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 5 | category | varchar(100) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 6 | subject | varchar(255) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 7 | message | text | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 8 | status | enum('Pending', 'Resolved') | utf8mb4_general_ci | | No | Pending | | | Change Drop More |
| 9 | created_at | timestamp | | | No | current_timestamp() | | | Change Drop More |
| 10 | updated_at | timestamp | | | No | current_timestamp() | | ON UPDATE CURRENT_TIMESTAMP() | Change Drop More |

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | eventID | int(11) | | | No | None | | AUTO_INCREMENT | Change Drop More |
| 2 | organizerID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 3 | title | varchar(100) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 4 | description | text | utf8mb4_general_ci | | Yes | NULL | | | Change Drop More |
| 5 | date | datetime | | | No | None | | | Change Drop More |
| 6 | location | varchar(100) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 7 | capacity | int(11) | | | No | 50 | | | Change Drop More |
| 8 | department | varchar(100) | utf8mb4_general_ci | | No | All | | | Change Drop More |
| 9 | gender | enum('Male', 'Female', 'Both') | utf8mb4_general_ci | | No | Both | | | Change Drop More |
| 10 | is_paid | enum('Yes', 'No') | utf8mb4_general_ci | | No | No | | | Change Drop More |
| 11 | price | decimal(8,2) | | | Yes | 0.00 | | | Change Drop More |
| 12 | status | enum('pending', 'approved', 'rejected', 'cancelled... | utf8mb4_general_ci | | Yes | pending | | | Change Drop More |
| 13 | created_at | timestamp | | | No | current_timestamp() | | | Change Drop More |

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | approvalID | int(11) | | | No | None | | AUTO_INCREMENT | Change Drop More |
| 2 | eventID | int(11) | | | No | None | | | Change Drop More |
| 3 | adminID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 4 | decision | enum('approved', 'rejected') | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 5 | decisionDate | timestamp | | | No | current_timestamp() | | | Change Drop More |
| 6 | remarks | text | utf8mb4_general_ci | | Yes | NULL | | | Change Drop More |

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | registrationID | int(11) | | | No | None | | AUTO_INCREMENT | Change Drop More |
| 2 | studentID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 3 | eventID | int(11) | | | No | None | | | Change Drop More |
| 4 | registrationDate | timestamp | | | No | current_timestamp() | | | Change Drop More |
| 5 | status | enum('confirmed', 'cancelled') | utf8mb4_general_ci | | Yes | confirmed | | | Change Drop More |

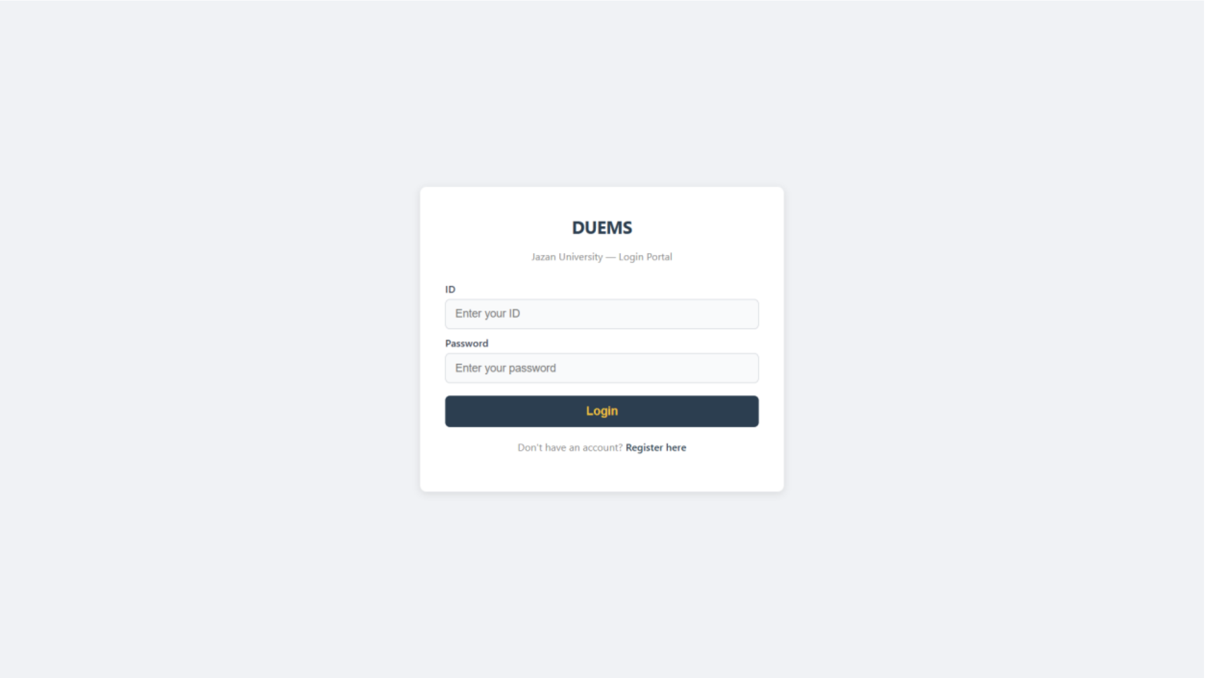
| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | studentID | varchar(9) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 2 | name | varchar(60) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 3 | email | varchar(80) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 4 | password | varchar(255) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 5 | department | varchar(50) | utf8mb4_general_ci | | No | None | | | Change Drop More |
| 6 | gender | enum('Male', 'Female') | utf8mb4_general_ci | | No | None | | | Change Drop More |

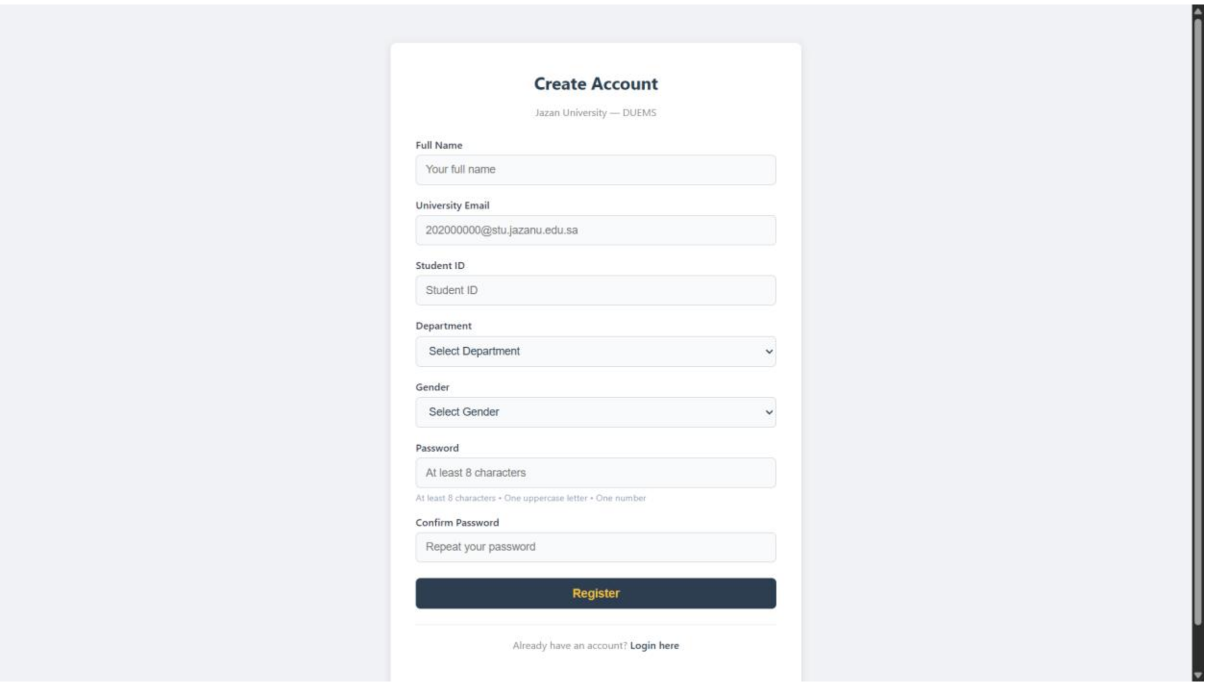



| Table | Action | Rows | Type | Collation | Size | Overhead |
|---|---|---|---|---|---|---|
| admins | Browse Structure Search Insert Empty Drop | 1 | InnoDB | utf8mb4_general_ci | 32.0 KiB | - |
| approvals | Browse Structure Search Insert Empty Drop | 3 | InnoDB | utf8mb4_general_ci | 48.0 KiB | - |
| events | Browse Structure Search Insert Empty Drop | 4 | InnoDB | utf8mb4_general_ci | 32.0 KiB | - |
| organizers | Browse Structure Search Insert Empty Drop | 1 | InnoDB | utf8mb4_general_ci | 32.0 KiB | - |
| registrations | Browse Structure Search Insert Empty Drop | 4 | InnoDB | utf8mb4_general_ci | 48.0 KiB | - |
| requests | Browse Structure Search Insert Empty Drop | 3 | InnoDB | utf8mb4_general_ci | 48.0 KiB | - |
| responses | Browse Structure Search Insert Empty Drop | 2 | InnoDB | utf8mb4_general_ci | 48.0 KiB | - |
| students | Browse Structure Search Insert Empty Drop | 4 | InnoDB | utf8mb4_general_ci | 32.0 KiB | - |

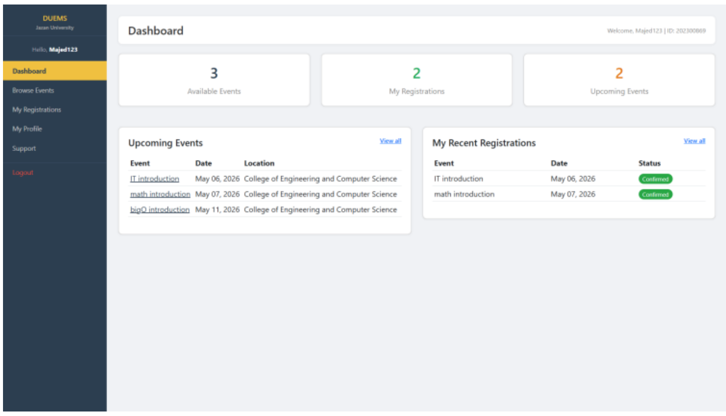
# 7. User Interface Design:

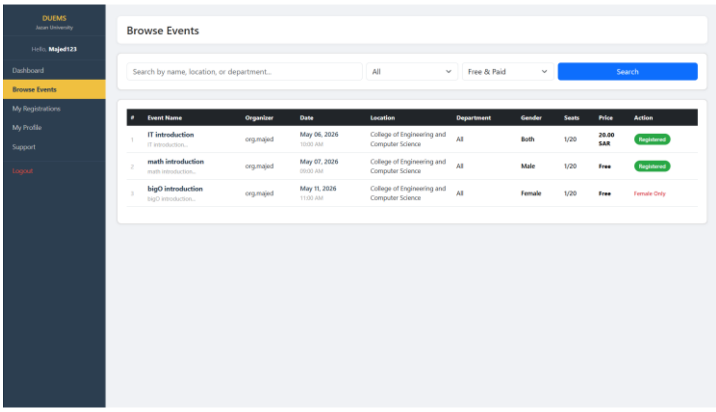
## Login pages

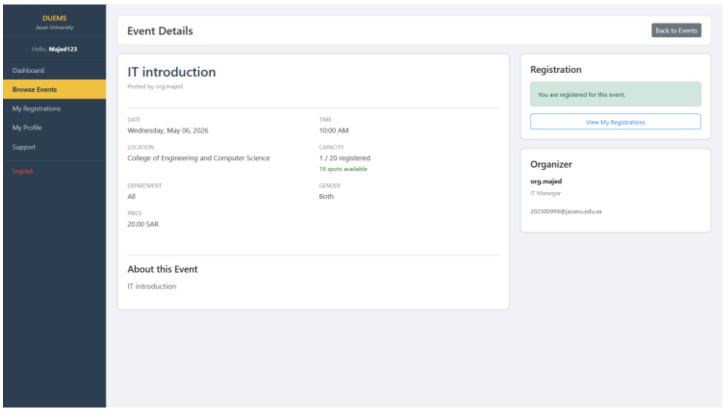


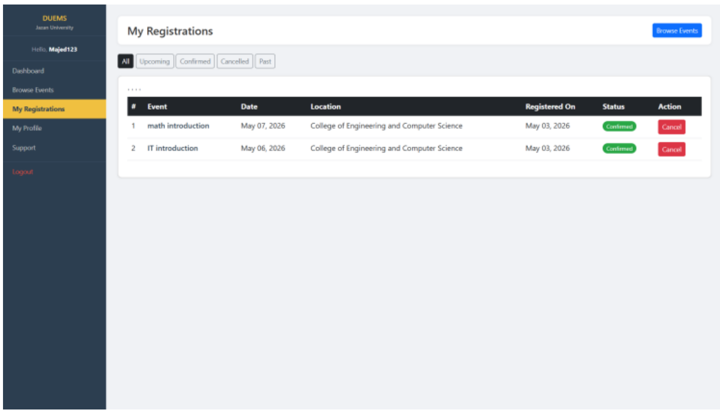



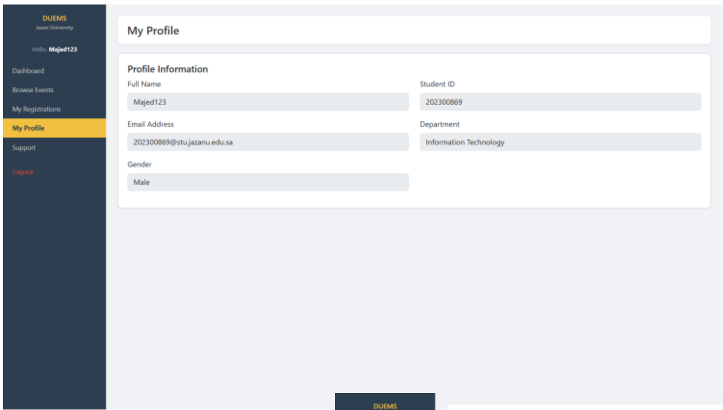
# Student pages

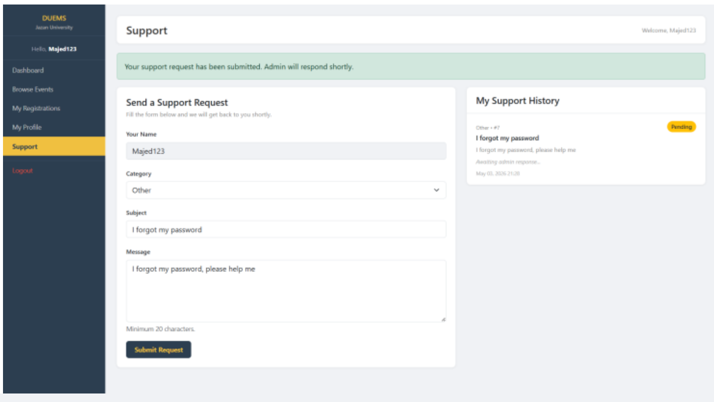


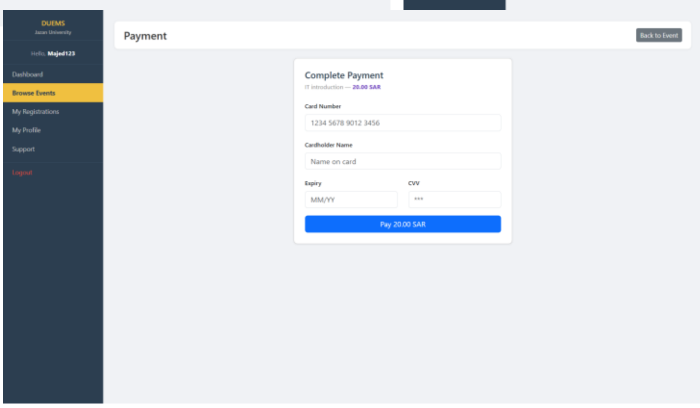



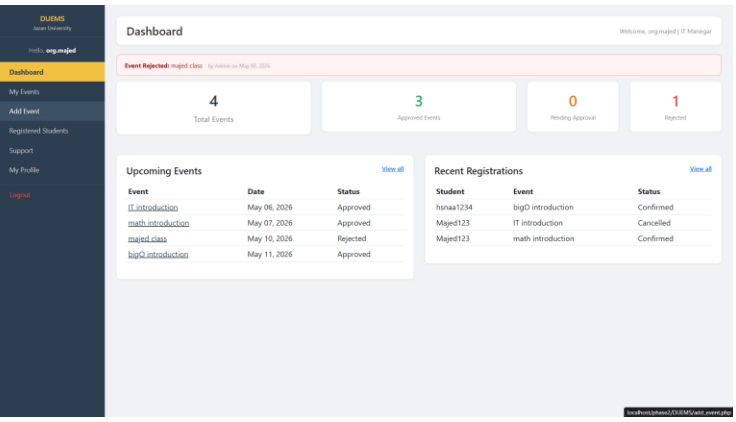



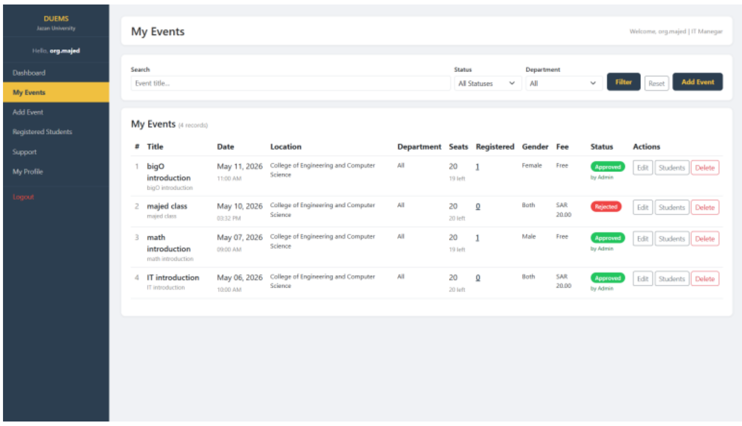



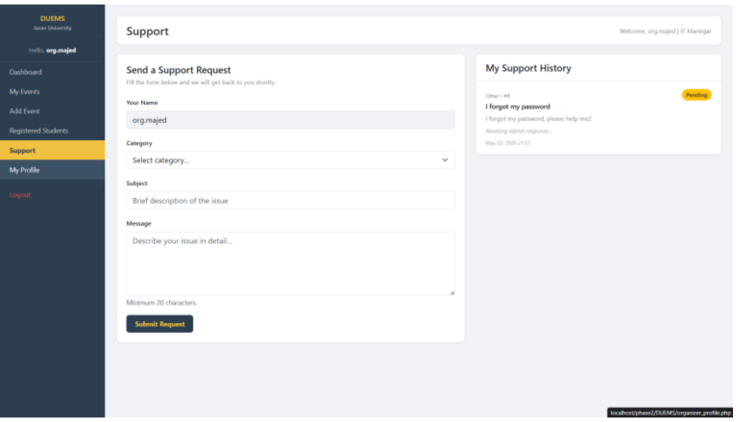



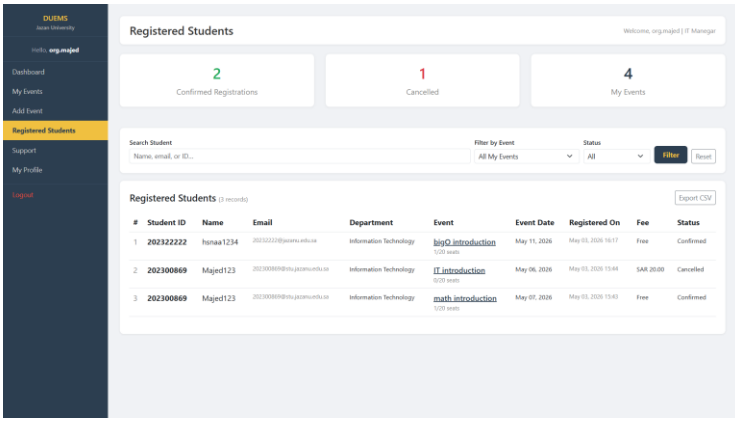



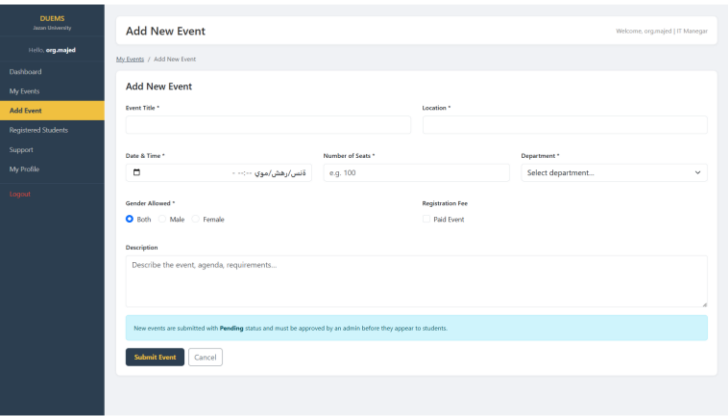
# Organizer pages

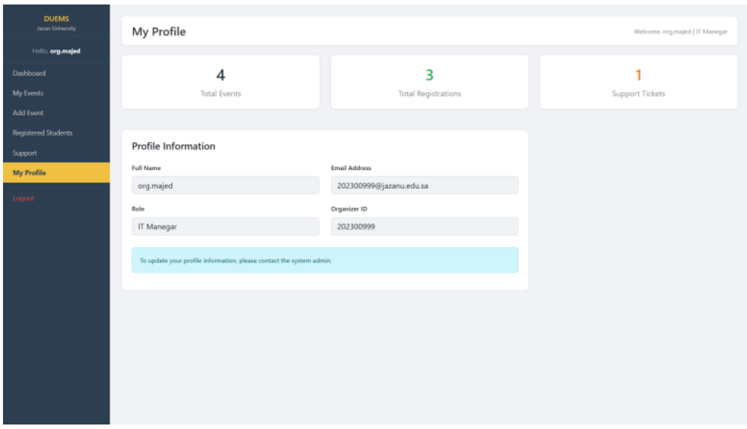


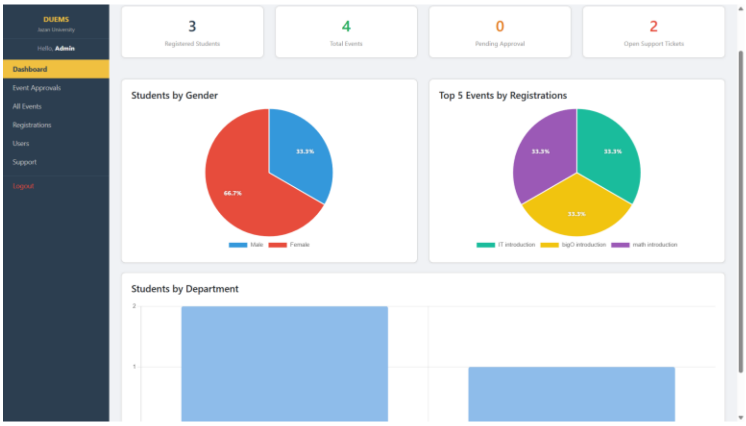



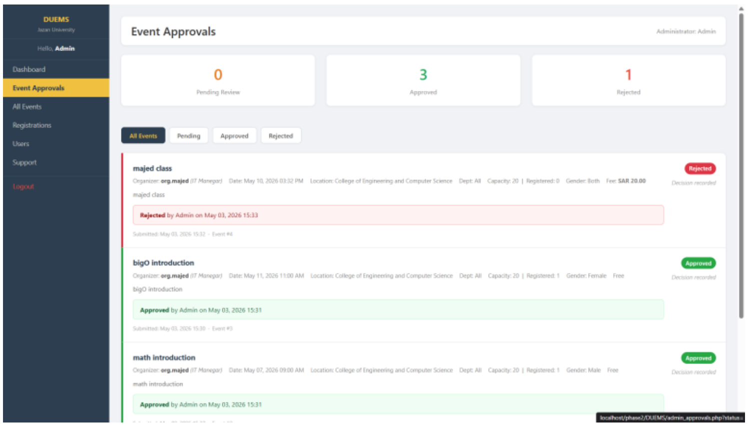



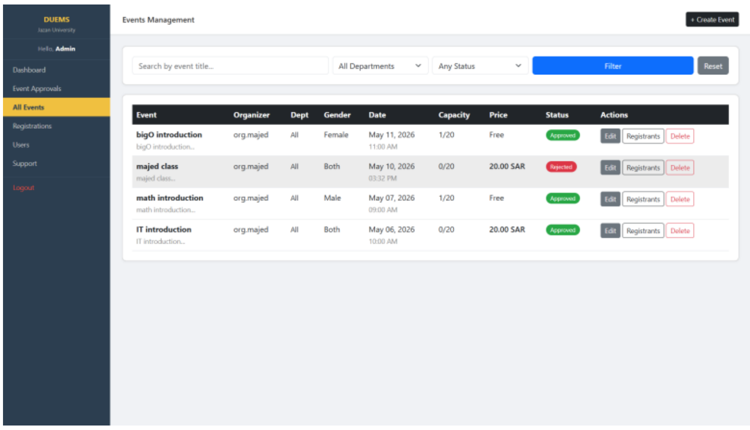



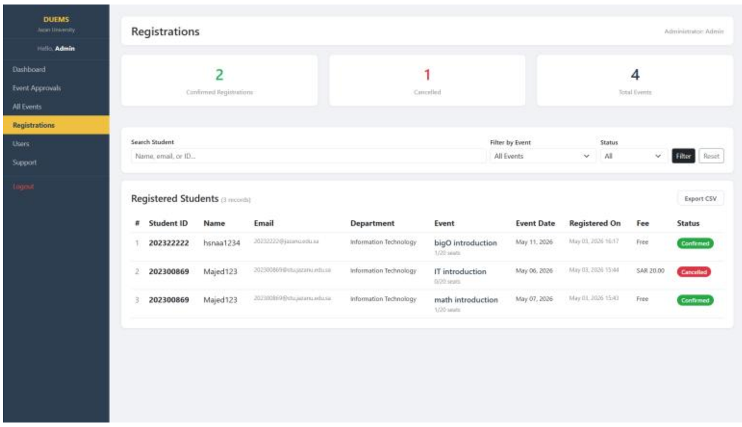



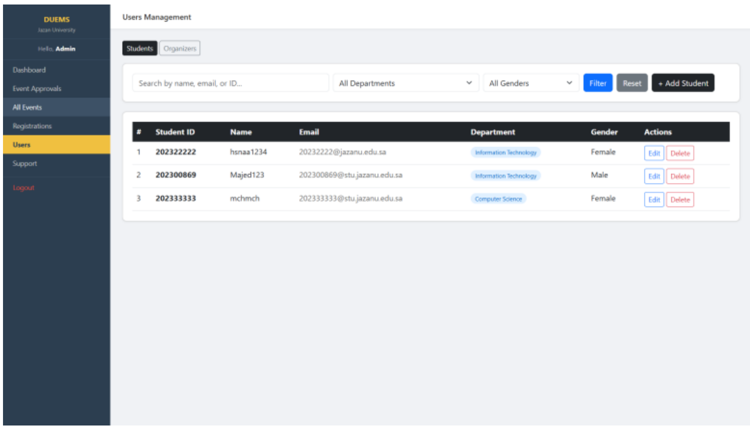
# Admin pages

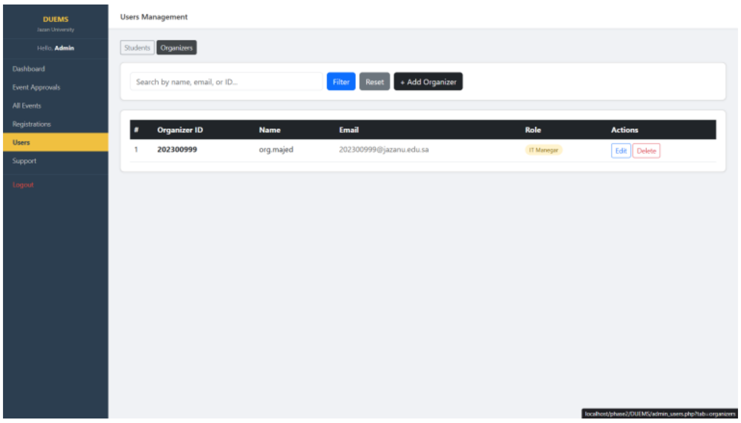

DUEMS
Dashboard
Event Approvals
All Events
Registrations
Users
Support
Logout
Students by Gender
Top 5 Events by Registrations
Students by Department

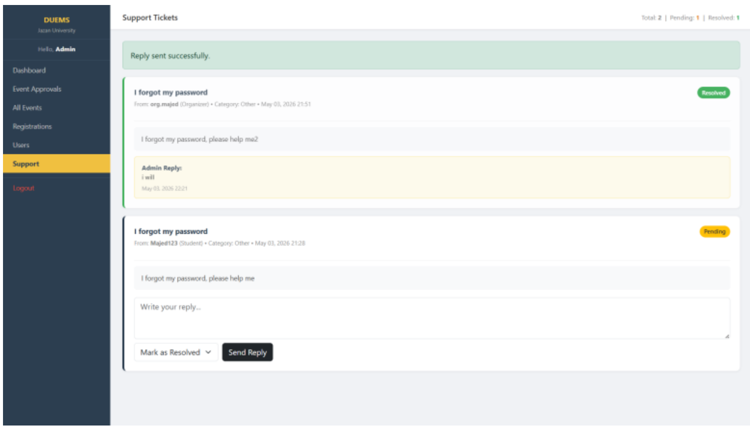

Event Approvals
majed class
bigO introduction
math introduction


Events Management
+ Create Event
Filter
Reset
bigO introduction
majed class
math introduction
IT introduction


Registrations
Registered Students
Export CSV

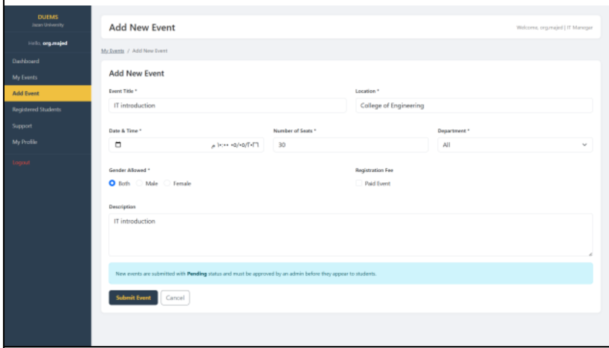
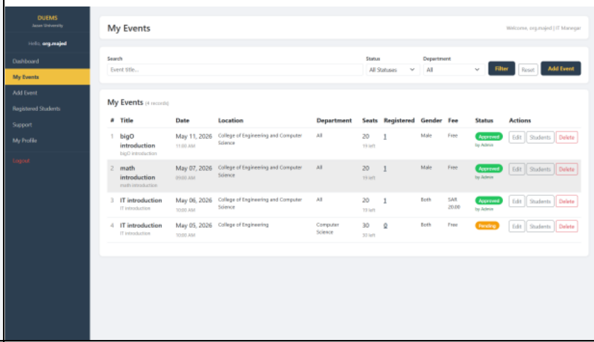
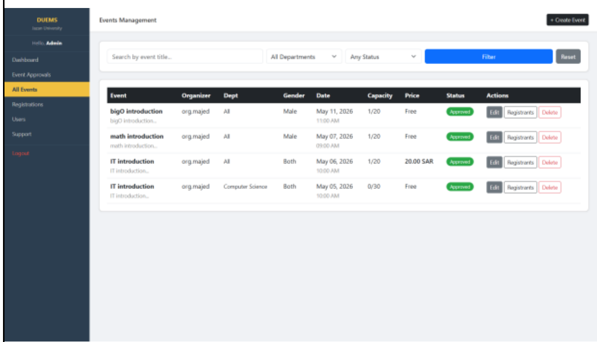
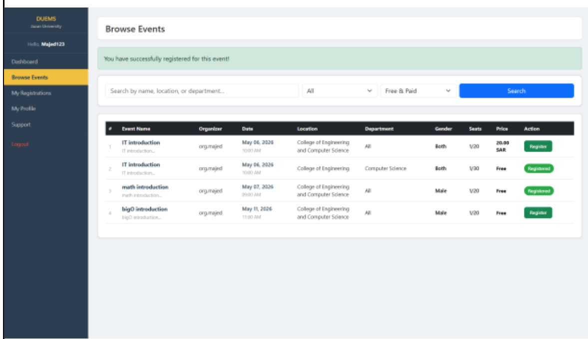

Users Management
Filter
Reset
+ Add Student


Users Management
Filter
Reset
+ Add Organizer


Support Tickets
Reply sent successfully.
I forgot my password
I forgot my password
Send Reply

# 8. Real Screen Shots of Process

## 8.1 Real Screen Shots Event Process

| | |
|---|---|
| 1.Create Event form organizer  | 2.Event is pending state wait the admin to accept the event  |
| 3.admin accept the event  | 4.The event is show for students and the student requester in the event  |

# 9. Coding of Important Functions and Procedures of Software

1. Code to make connection to Database:

```
<?php
$host     = 'localhost';
$dbname   = 'duems';
$username = 'root';
$password = '';
$conn = mysqli_connect($host, $username, $password, $dbname);
if (!$conn) {
die('Database connection failed: ' . mysqli_connect_error());
}
?>
```

2. HTML for Login page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Login - DUEMS</title>
  <style>
//ccs code
  </style>
</head>
<body>
<div class="card">
  <h2>DUEMS</h2>
  <p class="sub">Jazan University &mdash; Login Portal</p>
  <?php if ($error): ?>
    <div class="alert-error"><?= htmlspecialchars($error) ?></div>
  <?php endif; ?>
  <form method="POST" action="login.php" novalidate>
    <label for="identifier">ID </label>
    <input type="text" id="identifier" name="identifier" maxlength="9"
        placeholder="Enter your ID"
        value="<?= isset($_POST['identifier']) ? htmlspecialchars($_POST['identifier']) : '' ?>"
        autocomplete="username">
    <label for="password">Password</label>
    <input type="password" id="password" name="password"
        placeholder="Enter your password"
        autocomplete="current-password">


    <button type="submit">Login</button>
  </form>
  <p class="register-link">
    Don't have an account? <a href="register.php">Register here</a>
  </p>
</div>
</body></html>
```

3. PHP for login page

```
<?php session_start();
if (isset($_SESSION['student_id']))   { header('Location: student_dashboard.php');   exit; }
if (isset($_SESSION['organizer_id'])) { header('Location: organizer_dashboard.php'); exit; }
if (isset($_SESSION['admin_id']))     { header('Location: admin_dashboard.php');     exit; }
require_once 'db.php';
$error = '';
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
  $identifier = trim($_POST['identifier'] ?? '');
  $password   = trim($_POST['password']   ?? '');
  if (empty($identifier) || empty($password)) {
    $error = 'Please fill in all fields.';} else {
    $found = false;
    // 1. Student: always look up by studentID
    $stmt = mysqli_prepare($conn, "SELECT studentID, name, password FROM students WHERE studentID = ?");
    mysqli_stmt_bind_param($stmt, 's', $identifier);
    mysqli_stmt_execute($stmt);
    $user = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));
    mysqli_stmt_close($stmt);
    if ($user) {
      $found = true;
      if (!password_verify($password, $user['password'])) {
        $error = 'Wrong password.';
      } else {
        $_SESSION['student_id']   = $user['studentID'];
        $_SESSION['student_name'] = $user['name'];
        header('Location: student_dashboard.php'); exit;}}
    if (!$found) {
      // If not found by email and identifier is a short integer, try by organizerID
      if (!$user && ctype_digit($identifier) && strlen($identifier) <= 9) {
        $oid  = (int)$identifier;
        $stmt = mysqli_prepare($conn, "SELECT organizerID, name, password, role FROM organizers WHERE organizerID = ?");
        mysqli_stmt_bind_param($stmt, 'i', $oid);
```

```
        mysqli_stmt_execute($stmt);
        $user = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));
        mysqli_stmt_close($stmt);}
      if ($user) {
        $found = true;
        if (!password_verify($password, $user['password'])) {
          $error = 'Wrong password.';
        } else {
          $_SESSION['organizer_id']   = $user['organizerID'];
          $_SESSION['organizer_name'] = $user['name'];
          $_SESSION['organizer_role'] = $user['role'];
          header('Location: organizer_dashboard.php'); exit;}}}
    // 3. Admin: try email first, then short numeric ID
    if (!$found) {
      // If not found by email and short integer, try by adminID
      if (!$user && ctype_digit($identifier) && strlen($identifier) <= 9) {
        $aid  = (int)$identifier;
        $stmt = mysqli_prepare($conn, "SELECT adminID, name, password FROM admins
WHERE adminID = ?");
        mysqli_stmt_bind_param($stmt, 'i', $aid);
        mysqli_stmt_execute($stmt);
        $user = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));
        mysqli_stmt_close($stmt);}
      if ($user) {
        $found = true;
        if (!password_verify($password, $user['password'])) {
          $error = 'Wrong password.';
        } else {
          $_SESSION['admin_id']   = $user['adminID'];
          $_SESSION['admin_name'] = $user['name'];
          header('Location: admin_dashboard.php'); exit;}}}
    if (!$found) {
      $error = 'Account not found. Check your ID or email.';
    }
  }}?>
```

4. HTML for requester page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Register - DUEMS</title>
  <style>
//css code
  </style>
</head>
<body>

<div class="card">
  <h2>Create Account</h2>
  <p class="sub">Jazan University &mdash; DUEMS</p>

  <?php if ($error): ?>
    <div class="alert-error"><?= htmlspecialchars($error) ?></div>
  <?php endif; ?>

  <?php if ($success): ?>
    <div class="alert-success">
      <?= htmlspecialchars($success) ?>  <a href="login.php">Login now &rarr;</a>
    </div>
  <?php endif; ?>
  <!-- novalidate disables the browser's built-in popup -->
  <form method="POST" action="register.php" novalidate>
    <label>Full Name</label>
    <input type="text" name="name" placeholder="Your full name"
        value="<?= htmlspecialchars($_POST['name'] ?? '') ?>">
    <label>University Email</label>
    <input type="email" name="email" placeholder="202000000@stu.jazanu.edu.sa"
        value="<?= htmlspecialchars($_POST['email'] ?? '') ?>">
    <label>Student ID</label>
    <input type="text" name="student_id" placeholder="Student ID" maxlength="9"
```

```
      value="<?= htmlspecialchars($_POST['student_id'] ?? '') ?>">
    <label>Department</label>
    <select name="department">
      <option value=""> Select Department </option>
      <?php
      $depts = ['Computer Science','Information Technology','Software Engineering',
            'Electrical Engineering','Mechanical Engineering','Civil Engineering',
            'Business Administration','Medicine','Pharmacy','Other'];
      foreach ($depts as $d):
        $sel = (isset($_POST['department']) && $_POST['department'] == $d) ? 'selected' : '';
      ?>
        <option value="<?= $d ?>" <?= $sel ?>><?= $d ?></option>
      <?php endforeach; ?>
    </select>
    <label>Gender</label>
    <select name="gender">
      <option value=""> Select Gender </option>
      <option value="Male"   <?= (isset($_POST['gender']) && $_POST['gender'] == 'Male')
? 'selected' : '' ?>>Male</option>
      <option value="Female" <?= (isset($_POST['gender']) && $_POST['gender'] ==
'Female') ? 'selected' : '' ?>>Female</option>
    </select>
    <label>Password</label>
    <input type="password" name="password" placeholder="At least 8 characters">
    <p class="hint">At least 8 characters &bull; One uppercase letter &bull; One
number</p>
    <label>Confirm Password</label>
    <input type="password" name="confirm_password" placeholder="Repeat your
password">
    <button type="submit">Register</button>
  </form>
  <hr>
  <p class="login-link">Already have an account? <a href="login.php">Login
here</a></p>
</div>
</body>
</html>
```

5. PHP code for requester

```php
<?php
session_start();

if (isset($_SESSION['student_id'])) {
    header('Location: student_dashboard.php');
    exit;
}
require_once 'db.php';
$error   = '';
$success = '';
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $student_id = trim($_POST['student_id']);
    $name       = trim($_POST['name']);
    $email      = trim($_POST['email']);
    $password   = $_POST['password'];
    $confirm    = $_POST['confirm_password'];
    $department = trim($_POST['department']);
    $gender     = trim($_POST['gender'] ?? '');
    if (empty($name)) {
        $error = 'Please enter your full name.';
    } elseif (empty($email)) {
        $error = 'Please enter your university email.';
    } elseif (!str_ends_with($email, '@stu.jazanu.edu.sa')) {
        $error = 'Email must be a university email (@stu.jazanu.edu.sa).';
    } elseif (empty($student_id)) {
        $error = 'Please enter your Student ID.';
    } elseif (!preg_match('/^[0-9]{9}$/', $student_id)) {
        $error = 'Student ID must be exactly 9 digits.';
    } elseif (empty($department)) {
        $error = 'Please select your department.';
    } elseif (empty($gender) || !in_array($gender, ['Male', 'Female'])) {
        $error = 'Please select your gender.';
    } elseif (empty($password)) {
        $error = 'Please enter a password.';
```

```
    } elseif (strlen($password) < 8) {
        $error = 'Password must be at least 8 characters.';
    } elseif (!preg_match('/[A-Z]/', $password)) {
        $error = 'Password must contain at least one uppercase letter.';
    } elseif (!preg_match('/[0-9]/', $password)) {
        $error = 'Password must contain at least one number.';
    } elseif (empty($confirm)) {
        $error = 'Please confirm your password.';
    } elseif ($password !== $confirm) {
        $error = 'Passwords do not match.';
    } else {
        $stmt = mysqli_prepare($conn, "SELECT studentID FROM students WHERE studentID = ?");
        mysqli_stmt_bind_param($stmt, 's', $student_id);
        mysqli_stmt_execute($stmt);
        mysqli_stmt_store_result($stmt);
        if (mysqli_stmt_num_rows($stmt) > 0) {
            $error = 'This Student ID is already registered.';}
        mysqli_stmt_close($stmt);
        if (!$error) {
            $stmt = mysqli_prepare($conn, "SELECT studentID FROM students WHERE email = ?");
            mysqli_stmt_bind_param($stmt, 's', $email);
            mysqli_stmt_execute($stmt);
            mysqli_stmt_store_result($stmt);
            if (mysqli_stmt_num_rows($stmt) > 0) {
                $error = 'This email is already registered.';}
            mysqli_stmt_close($stmt);}if (!$error) {
            $hashed = password_hash($password, PASSWORD_BCRYPT);
            $sql    = "INSERT INTO students (studentID, name, email, password, department, gender) VALUES (?, ?, ?, ?, ?, ?)";
            $stmt   = mysqli_prepare($conn, $sql);
            mysqli_stmt_bind_param($stmt, 'ssssss', $student_id, $name, $email, $hashed, $department, $gender);
            if (mysqli_stmt_execute($stmt)) {
                $success = 'Account created successfully! You can now log in.';} else {$error = 'Something went wrong. Please try again.';}mysqli_stmt_close($stmt);}}}?>
```

6. PHP code for add event

```php
<?php
session_start();
if (!isset($_SESSION['organizer_id'])) { header('Location: login.php'); exit; }
require_once 'db.php';
$organizerID = (int)$_SESSION['organizer_id'];
$row = mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT name, role FROM organizers WHERE organizerID = $organizerID"));
$organizerName = $row['name'] ?? 'Organizer';
$organizerRole = $row['role'] ?? '';
$_SESSION['organizer_name'] = $organizerName;
$errors = [];
$departments = [
    'All','Computer Science','Engineering','Business Administration',
    'Medicine','Pharmacy','Education','Sciences','Arts & Humanities',
    'Law','Architecture','Nursing','Dentistry'
];
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $title       = trim($_POST['title']       ?? '');
    $description = trim($_POST['description'] ?? '');
    $date        = trim($_POST['date']        ?? '');
    $location    = trim($_POST['location']    ?? '');
    $capacity    = (int)($_POST['capacity']   ?? 0);
    $department  = trim($_POST['department']  ?? '');
    $gender      = trim($_POST['gender']      ?? 'Both');
    $is_paid     = isset($_POST['is_paid'])   ? 'Yes' : 'No';
    $price       = ($is_paid === 'Yes')       ? (float)($_POST['price'] ?? 0) : 0.00;

    if (!$title)       $errors[] = 'Event title is required.';
    if (!$date)        $errors[] = 'Date & time is required.';
    if (!$location)    $errors[] = 'Location is required.';
    if ($capacity < 1) $errors[] = 'Capacity must be at least 1.';
    if (!$department)  $errors[] = 'Department is required.';
    if (!in_array($gender, ['Male','Female','Both'])) $errors[] = 'Invalid gender selection.';
    if ($is_paid === 'Yes' && $price <= 0) $errors[] = 'Please enter a valid price for paid events.';
```

```
    if (empty($errors)) {
        $stmt = mysqli_prepare($conn,
            "INSERT INTO events (organizerID, title, description, date, location, capacity, department, gender, is_paid, price, status)
             VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, 'pending')");
        mysqli_stmt_bind_param($stmt, 'issssisssd',
            $organizerID, $title, $description, $date, $location, $capacity, $department, $gender, $is_paid, $price);
        if (mysqli_stmt_execute($stmt)) { header('Location: events.php?msg=added'); exit; }
        else $errors[] = 'Database error: ' . mysqli_error($conn);
    }
}
$current_page = 'add_event.php';
$page_title   = 'Add New Event';
?>
```

7. PHP code for admin approval for event

```
<?php
session_start();
if (!isset($_SESSION['admin_id'])) { header('Location: login.php'); exit; }
require_once 'db.php';
$admin_id   = (int)$_SESSION['admin_id'];
$admin_name = $_SESSION['admin_name'] ?? 'Admin';
$msg        = '';
$msg_type   = '';
// Handle Approve / Reject
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action'])) {
    $action   = $_POST['action'];
    $event_id = (int)($_POST['event_id'] ?? 0);
    $remarks  = trim($_POST['remarks']   ?? '');
    if ($event_id && in_array($action, ['approve', 'reject'])) {
        $new_status = ($action === 'approve') ? 'approved' : 'rejected';
        $decision   = $new_status;
        $stmt = mysqli_prepare($conn, "UPDATE events SET status = ? WHERE eventID = ?");
        mysqli_stmt_bind_param($stmt, 'si', $new_status, $event_id);
        mysqli_stmt_execute($stmt);
```

```
    mysqli_stmt_close($stmt);
    $stmt2 = mysqli_prepare($conn,
      "INSERT INTO approvals (eventID, adminID, decision, decisionDate, remarks)
       VALUES (?, ?, ?, NOW(), ?)");
    mysqli_stmt_bind_param($stmt2, 'iiss', $event_id, $admin_id, $decision, $remarks);
    mysqli_stmt_execute($stmt2);
    mysqli_stmt_close($stmt2);
    $msg      = $action === 'approve' ? 'Event approved successfully.' : 'Event rejected.';
    $msg_type = $action === 'approve' ? 'success' : 'danger';}}
// Stats
$cnt_pending  = (int)mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*) AS c
FROM events WHERE status='pending'"))['c'];
$cnt_approved = (int)mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*) AS c
FROM events WHERE status='approved'"))['c'];
$cnt_rejected = (int)mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*) AS c
FROM events WHERE status='rejected'"))['c'];
// Filter
$fStatus     = isset($_GET['status']) ? mysqli_real_escape_string($conn, $_GET['status']) :
'pending';
$whereStatus = $fStatus !== '' ? "AND e.status = '$fStatus'" : '';
// Fetch events
$events = mysqli_query($conn, "
  SELECT e.*,
       o.name AS org_name,
       o.role AS org_role,
       (SELECT COUNT(*) FROM registrations r WHERE r.eventID = e.eventID AND
r.status = 'confirmed') AS reg_count,
       ap.decision     AS ap_decision,
       ap.decisionDate AS ap_date,
       ap.remarks      AS ap_remarks,
       a.name          AS ap_admin_name
  FROM events e
  LEFT JOIN organizers o ON e.organizerID = o.organizerID
  LEFT JOIN approvals  ap ON ap.approvalID = (
    SELECT MAX(approvalID) FROM approvals WHERE eventID = e.eventID)
  LEFT JOIN admins a ON ap.adminID = a.adminIDWHERE 1=1 $whereStatusORDER BY
e.created_at DESC");?>
```

8. PHP for Admin overview

```
<?php
session_start();
if (!isset($_SESSION['admin_id'])) { header('Location: login.php'); exit; }
require_once 'db.php';
// Gender stats
$gender_stats = [];
$res = mysqli_query($conn, "SELECT gender, COUNT(*) as cnt FROM students WHERE
gender IN('Male','Female') GROUP BY gender");
while ($row = mysqli_fetch_assoc($res)) $gender_stats[$row['gender']] = (int)$row['cnt'];
$maleCount   = $gender_stats['Male']   ?? 0;
$femaleCount = $gender_stats['Female'] ?? 0;
// Department stats
$dept_names  = [];
$dept_counts = [];
$res = mysqli_query($conn, "SELECT department, COUNT(*) as cnt FROM students
GROUP BY department ORDER BY cnt DESC");
while ($row = mysqli_fetch_assoc($res)) {
  $dept_names[]  = $row['department'];
  $dept_counts[] = (int)$row['cnt'];}
// Top events by registrations
$event_names  = [];
$event_counts = [];
$res = mysqli_query($conn, "SELECT e.title, COUNT(r.registrationID) as cnt FROM events e
LEFT JOIN registrations r ON e.eventID = r.eventID GROUP BY e.eventID HAVING cnt > 0
ORDER BY cnt DESC LIMIT 5");
while ($row = mysqli_fetch_assoc($res)) {
  $event_names[]  = $row['title'];
  $event_counts[] = (int)$row['cnt'];}
// Summary stats
$total_students   = $maleCount + $femaleCount;
$total_events     = (int)mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*) as c
FROM events"))['c'];
$pending_events   = (int)mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*) as c
FROM events WHERE status='pending'"))['c'];
$pending_requests = (int)mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*) as c
FROM requests WHERE status='Pending'"))['c'];?>
```

```
<script>
// Shared percentage label plugin for pie charts
var pctPlugin = {
  id: 'pctLabels',
  afterDatasetDraw: function(chart) {
    var ctx = chart.ctx;
    var dataset = chart.data.datasets[0];
    var total = dataset.data.reduce(function(a, b) { return a + b; }, 0);
    chart.getDatasetMeta(0).data.forEach(function(arc, i) {
      var pct = total > 0 ? ((dataset.data[i] / total) * 100).toFixed(1) : 0;
      if (pct < 3) return; // skip tiny slices
      var mid = (arc.startAngle + arc.endAngle) / 2;
      var r   = (arc.outerRadius + arc.innerRadius) / 2 + (arc.outerRadius - arc.innerRadius)
* 0.15;
      var x   = arc.x + Math.cos(mid) * r;
      var y   = arc.y + Math.sin(mid) * r;
      ctx.save();
      ctx.fillStyle    = 'white';
      ctx.font         = 'bold 13px Segoe UI, sans-serif';
      ctx.textAlign    = 'center';
      ctx.textBaseline = 'middle';
      ctx.shadowColor  = 'rgba(0,0,0,0.35)';
      ctx.shadowBlur   = 3;
      ctx.fillText(pct + '%', x, y);
      ctx.restore();
    });
  }
};
<?php if ($maleCount + $femaleCount > 0): ?>
new Chart(document.getElementById('genderChart').getContext('2d'), {
  type: 'pie',
  plugins: [pctPlugin],
  data: {
    labels: ['Male', 'Female'],
    datasets: [{
```

```
            data: [<?= $maleCount ?>, <?= $femaleCount ?>],
            backgroundColor: ['#3498db', '#e74c3c'],
            borderWidth: 2,
            borderColor: '#fff'
        }]
    },
    options: {
        responsive: true,
        maintainAspectRatio: false,
        plugins: {
            legend: { position: 'bottom' },
            tooltip: {
                callbacks: {
                    label: function(ctx) {
                        var total = ctx.dataset.data.reduce(function(a,b){return a+b;},0);
                        var pct   = total > 0 ? ((ctx.parsed / total) * 100).toFixed(1) : 0;
                        return '  ' + ctx.label + ': ' + ctx.parsed + ' (' + pct + '%)';}}}}}});
<?php endif; ?>

<?php if (count($event_names) > 0): ?>
new Chart(document.getElementById('eventsChart').getContext('2d'), {
    type: 'pie',
    plugins: [pctPlugin],
    data: {
        labels: <?= json_encode($event_names) ?>,
        datasets: [{
            data: <?= json_encode($event_counts) ?>,
            backgroundColor: ['#1abc9c', '#f1c40f', '#9b59b6', '#34495e', '#e67e22'],
            borderWidth: 2,
            borderColor: '#fff'}]},



    options: {
        responsive: true,
```

```
        maintainAspectRatio: false,
        plugins: {
          legend: { position: 'bottom' },
          tooltip: {
            callbacks: {
              label: function(ctx) {
                var total = ctx.dataset.data.reduce(function(a,b){return a+b;},0);
                var pct   = total > 0 ? ((ctx.parsed / total) * 100).toFixed(1) : 0;
                return '  ' + ctx.label + ': ' + ctx.parsed + ' registrations (' + pct + '%)';}}}}}});
<?php endif; ?>
<?php if (count($dept_names) > 0): ?>
new Chart(document.getElementById('deptChart').getContext('2d'), {
    type: 'bar',
    data: {
        labels: <?= json_encode($dept_names) ?>,
        datasets: [{
          label: 'Students',
          data: <?= json_encode($dept_counts) ?>,
          backgroundColor: '#8ebcea',
          borderRadius: 4}]},
    options: {
        responsive: true,
        maintainAspectRatio: false,
        plugins: {
          legend: { display: false },
          tooltip: {
            callbacks: {
              label: function(ctx) {
                var total = ctx.dataset.data.reduce(function(a,b){return a+b;},0);
                var pct   = total > 0 ? ((ctx.parsed.y / total) * 100).toFixed(1) : 0;
                return '  Students: ' + ctx.parsed.y + ' (' + pct + '%)';}}}},
        scales: { y: { beginAtZero: true, ticks: { stepSize: 1 } } }}});<?php endif;
?></script></body></html><?php mysqli_close($conn); ?>
```

9. PHP for admin user

```
<?php
session_start();
if (!isset($_SESSION['admin_id'])) { header('Location: login.php'); exit; }
require_once 'db.php';

$all_depts = [
  'Computer Science','Information Technology','Software Engineering',
  'Electrical Engineering','Mechanical Engineering','Civil Engineering',
  'Business Administration','Medicine','Pharmacy','Other'
];

$msg      = '';
$msg_type = '';
$reopen_modal = '';
$reopen_data  = [];

//   POST Handler
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  $action = $_POST['action'] ?? '';

  // ADD STUDENT
  if ($action === 'add_student') {
    $sid   = trim($_POST['student_id']   ?? '');
    $name  = trim($_POST['s_name']       ?? '');
    $email = trim($_POST['s_email']      ?? '');
    $dept  = trim($_POST['s_department'] ?? '');
    $gen   = trim($_POST['s_gender']     ?? '');
    $pw    = $_POST['s_password']        ?? '';

    // Basic validation
    if ($sid === '' || $name === '' || $email === '' || $dept === '' || $gen === '' || $pw
=== '') {
      $msg = 'All fields are required.';
      $msg_type = 'danger';
```

```php
            $reopen_modal = 'add_student';
            $reopen_data  = compact('sid','name','email','dept','gen');
        } else {
            // Check if ID already exists BEFORE inserting
            $chk = mysqli_prepare($conn, "SELECT studentID FROM students WHERE studentID
= ?");
            mysqli_stmt_bind_param($chk, 's', $sid);
            mysqli_stmt_execute($chk);
            mysqli_stmt_store_result($chk);
            $exists = mysqli_stmt_num_rows($chk) > 0;
            mysqli_stmt_close($chk);

            if ($exists) {
                $msg = "Student ID \"$sid\" already exists in the database. Use a different ID.";
                $msg_type = 'danger';
                $reopen_modal = 'add_student';
                $reopen_data  = compact('sid','name','email','dept','gen');
            } else {
                $hash = password_hash($pw, PASSWORD_BCRYPT);
                $ins  = mysqli_prepare($conn,
                    "INSERT INTO students (studentID, name, email, password, department, gender)
                     VALUES (?, ?, ?, ?, ?, ?)");
                mysqli_stmt_bind_param($ins, 'ssssss', $sid, $name, $email, $hash, $dept, $gen);
                try {
                    mysqli_stmt_execute($ins);
                    mysqli_stmt_close($ins);
                    header("Location: admin_users.php?tab=students&msg=" .
                        urlencode("Student \"$name\" added successfully.") . "&msg_type=success");
                    exit;
                } catch (mysqli_sql_exception $e) {
                    mysqli_stmt_close($ins);
                    if ($e->getCode() == 1062) {
                        // Check which field is duplicated for a clear message
                        if (stripos($e->getMessage(), 'email') !== false) {
                            $msg = "Email \"$email\" is already registered to another student.";
```

```
                } else {
                    $msg = "Student ID \"$sid\" already exists. Please use a different ID.";
                }
            } else {
                $msg = "Database error: " . $e->getMessage();
            }
            $msg_type     = 'danger';
            $reopen_modal = 'add_student';
            $reopen_data  = compact('sid','name','email','dept','gen');
        }
    }
}
}

// EDIT STUDENT
elseif ($action === 'edit_student') {
    $sid   = trim($_POST['student_id']   ?? '');
    $name  = trim($_POST['s_name']       ?? '');
    $email = trim($_POST['s_email']      ?? '');
    $dept  = trim($_POST['s_department'] ?? '');
    $gen   = trim($_POST['s_gender']     ?? '');
    $pw    = $_POST['s_password']        ?? '';

    if (!empty($pw)) {
        $hash = password_hash($pw, PASSWORD_BCRYPT);
        $stmt = mysqli_prepare($conn,
            "UPDATE students SET name=?, email=?, password=?, department=?, gender=?
WHERE studentID=?");
        mysqli_stmt_bind_param($stmt, 'ssssss', $name, $email, $hash, $dept, $gen, $sid);
    } else {
        $stmt = mysqli_prepare($conn,
            "UPDATE students SET name=?, email=?, department=?, gender=? WHERE
studentID=?");
        mysqli_stmt_bind_param($stmt, 'sssss', $name, $email, $dept, $gen, $sid);
    }
```

```
        if (mysqli_stmt_execute($stmt)) {
            mysqli_stmt_close($stmt);
            header("Location: admin_users.php?tab=students&msg=" .
                   urlencode("Student updated successfully.") . "&msg_type=success");
            exit;
        } else {
            $err = mysqli_stmt_error($stmt);
            mysqli_stmt_close($stmt);
            $msg = "Update failed: $err";
            $msg_type = 'danger';
            $reopen_modal = 'edit_student';
            $reopen_data =
['studentID'=>$sid,'name'=>$name,'email'=>$email,'department'=>$dept,'gender'=>$gen];
        }
    }

    // ADD ORGANIZER
    elseif ($action === 'add_organizer') {
        $oid   = trim($_POST['o_id']       ?? '');
        $name  = trim($_POST['o_name']     ?? '');
        $email = trim($_POST['o_email']    ?? '');
        $role  = trim($_POST['o_role']     ?? '');
        $pw    = $_POST['o_password']      ?? '';

        if ($oid === '' || $name === '' || $email === '' || $role === '' || $pw === '') {
            $msg = 'All fields are required.';
            $msg_type = 'danger';
            $reopen_modal = 'add_org';
            $reopen_data  = compact('oid','name','email','role');
        } else {
            // Pre-check ID
            $chk = mysqli_prepare($conn, "SELECT organizerID FROM organizers WHERE
organizerID = ?");
            mysqli_stmt_bind_param($chk, 's', $oid);
            mysqli_stmt_execute($chk);
            mysqli_stmt_store_result($chk);
```

```
        $exists = mysqli_stmt_num_rows($chk) > 0;
        mysqli_stmt_close($chk);

        if ($exists) {
          $msg = "Organizer ID \"$oid\" already exists. Use a different ID.";
          $msg_type = 'danger';
          $reopen_modal = 'add_org';
          $reopen_data  = compact('oid','name','email','role');
        } else {
          $hash = password_hash($pw, PASSWORD_BCRYPT);
          $ins  = mysqli_prepare($conn,
            "INSERT INTO organizers (organizerID, name, email, password, role) VALUES (?,
?, ?, ?, ?)");
          mysqli_stmt_bind_param($ins, 'sssss', $oid, $name, $email, $hash, $role);
          try {
            mysqli_stmt_execute($ins);
            mysqli_stmt_close($ins);
            header("Location: admin_users.php?tab=organizers&msg=" .
              urlencode("Organizer \"$name\" added successfully.") .
"&msg_type=success");
            exit;
          } catch (mysqli_sql_exception $e) {
            mysqli_stmt_close($ins);
            if ($e->getCode() == 1062) {
              if (stripos($e->getMessage(), 'email') !== false) {
                $msg = "Email \"$email\" is already registered to another organizer.";
              } else {
                $msg = "Organizer ID \"$oid\" already exists. Please use a different ID.";
              }
            } else {
              $msg = "Database error: " . $e->getMessage();
            }
            $msg_type    = 'danger';
            $reopen_modal = 'add_org';
            $reopen_data  = compact('oid','name','email','role');
          }
```

```
        }
    }
}

// EDIT ORGANIZER
elseif ($action === 'edit_organizer') {
    $oid   = trim($_POST['o_id_hidden'] ?? '');
    $name  = trim($_POST['o_name']     ?? '');
    $email = trim($_POST['o_email']    ?? '');
    $role  = trim($_POST['o_role']     ?? '');
    $pw    = $_POST['o_password']      ?? '';

    if (!empty($pw)) {
        $hash = password_hash($pw, PASSWORD_BCRYPT);
        $stmt = mysqli_prepare($conn,
            "UPDATE organizers SET name=?, email=?, password=?, role=? WHERE
organizerID=?");
        mysqli_stmt_bind_param($stmt, 'sssss', $name, $email, $hash, $role, $oid);
    } else {
        $stmt = mysqli_prepare($conn,
            "UPDATE organizers SET name=?, email=?, role=? WHERE organizerID=?");
        mysqli_stmt_bind_param($stmt, 'ssss', $name, $email, $role, $oid);
    }

    if (mysqli_stmt_execute($stmt)) {
        mysqli_stmt_close($stmt);
        header("Location: admin_users.php?tab=organizers&msg=" .
              urlencode("Organizer updated successfully.") . "&msg_type=success");
        exit;
    } else {
        $err = mysqli_stmt_error($stmt);
        mysqli_stmt_close($stmt);
        $msg = "Update failed: $err";
        $msg_type = 'danger';
    }
```

```
    }

    // DELETE STUDENT
    elseif ($action === 'delete_student') {
      $sid = trim($_POST['student_id'] ?? '');
      $s1  = mysqli_prepare($conn, "DELETE FROM registrations WHERE studentID=?");
      mysqli_stmt_bind_param($s1, 's', $sid);
      mysqli_stmt_execute($s1);
      mysqli_stmt_close($s1);
      $s2  = mysqli_prepare($conn, "DELETE FROM students WHERE studentID=?");
      mysqli_stmt_bind_param($s2, 's', $sid);
      mysqli_stmt_execute($s2);
      mysqli_stmt_close($s2);
      header("Location: admin_users.php?tab=students&msg=" .
          urlencode("Student deleted.") . "&msg_type=success");
      exit;
    }

    // DELETE ORGANIZER
    elseif ($action === 'delete_organizer') {
      $oid = trim($_POST['organizer_id'] ?? '');
      $s1  = mysqli_prepare($conn, "DELETE FROM events WHERE organizerID=?");
      mysqli_stmt_bind_param($s1, 's', $oid);
      mysqli_stmt_execute($s1);
      mysqli_stmt_close($s1);
      $s2  = mysqli_prepare($conn, "DELETE FROM organizers WHERE organizerID=?");
      mysqli_stmt_bind_param($s2, 's', $oid);
      mysqli_stmt_execute($s2);
      mysqli_stmt_close($s2);
      header("Location: admin_users.php?tab=organizers&msg=" .
          urlencode("Organizer deleted.") . "&msg_type=success");
      exit;
    }
  }
```

```php
// GET message from redirect
if ($msg === '' && isset($_GET['msg'])) {
    $msg      = htmlspecialchars($_GET['msg']);
    $msg_type = $_GET['msg_type'] ?? 'success';
}

//   Page state
$active_tab = $_GET['tab'] ?? 'students';
$search     = isset($_GET['search']) ? mysqli_real_escape_string($conn, trim($_GET['search'])) : '';
$fDept      = isset($_GET['dept'])   ? mysqli_real_escape_string($conn, $_GET['dept'])         : '';
$fGender    = isset($_GET['gender']) ? mysqli_real_escape_string($conn, $_GET['gender'])       : '';

// Which modal to open
$show_add_student = isset($_GET['add_student'])  || $reopen_modal === 'add_student';
$show_add_org     = isset($_GET['add_org'])      || $reopen_modal === 'add_org';
$edit_student_id   = $_GET['edit_student']            ?? ($reopen_modal === 'edit_student' ? ($reopen_data['studentID'] ?? null) : null);
$edit_org_id       = $_GET['edit_org']           ?? null;
$confirm_del_sid  = $_GET['confirm_del_s']      ?? null;
$confirm_del_oid  = $_GET['confirm_del_o']      ?? null;

// Fetch edit rows
$edit_student_row = null;
if ($reopen_modal === 'edit_student') {
    $edit_student_row = $reopen_data;
} elseif ($edit_student_id) {
    $r = mysqli_prepare($conn, "SELECT * FROM students WHERE studentID=?");
    mysqli_stmt_bind_param($r, 's', $edit_student_id);
    mysqli_stmt_execute($r);
    $edit_student_row = mysqli_fetch_assoc(mysqli_stmt_get_result($r));
    mysqli_stmt_close($r);
}

$edit_org_row = null;
```

```
if ($edit_org_id) {
    $r = mysqli_prepare($conn, "SELECT * FROM organizers WHERE organizerID=?");
    mysqli_stmt_bind_param($r, 's', $edit_org_id);
    mysqli_stmt_execute($r);
    $edit_org_row = mysqli_fetch_assoc(mysqli_stmt_get_result($r));
    mysqli_stmt_close($r);
}

// Table data
if ($active_tab === 'students') {
    $w = "WHERE (name LIKE '%$search%' OR email LIKE '%$search%' OR studentID LIKE
'%$search%')";
    if ($fDept)   $w .= " AND department='$fDept'";
    if ($fGender) $w .= " AND gender='$fGender'";
    $data_res = mysqli_query($conn, "SELECT * FROM students $w ORDER BY name ASC");
} else {
    $w = "WHERE (name LIKE '%$search%' OR email LIKE '%$search%' OR organizerID LIKE
'%$search%')";
    $data_res = mysqli_query($conn, "SELECT * FROM organizers $w ORDER BY name ASC");
}
?>
```

10. PHP code for event detail

```php
<?php
session_start();
if (!isset($_SESSION['student_id'])) {
    header('Location: login.php');
    exit;
}
require_once 'db.php';

$student_id = $_SESSION['student_id'];
$message    = '';
$msg_type   = '';
// get event id from url
if (!isset($_GET['id']) || !is_numeric($_GET['id'])) {
    header('Location: browse_events.php');
    exit;
}
$event_id = (int)$_GET['id'];

// handle register button
if (isset($_POST['register_event'])) {
    $stmt = mysqli_prepare($conn, "SELECT registrationID FROM registrations WHERE studentID = ? AND eventID = ? AND status != 'cancelled'");
    mysqli_stmt_bind_param($stmt, 'si', $student_id, $event_id);
    mysqli_stmt_execute($stmt);
    mysqli_stmt_store_result($stmt);
    $already = mysqli_stmt_num_rows($stmt) > 0;
    mysqli_stmt_close($stmt);

    if ($already) {
        $message  = 'You are already registered for this event.';
        $msg_type = 'warning';
    } else {
        $stmt = mysqli_prepare($conn, "SELECT capacity FROM events WHERE eventID = ?");
        mysqli_stmt_bind_param($stmt, 'i', $event_id);
```

```
mysqli_stmt_execute($stmt);
$res = mysqli_stmt_get_result($stmt);
$ev  = mysqli_fetch_assoc($res);
mysqli_stmt_close($stmt);

$stmt = mysqli_prepare($conn, "SELECT COUNT(*) AS cnt FROM registrations WHERE eventID = ? AND status = 'confirmed'");
mysqli_stmt_bind_param($stmt, 'i', $event_id);
mysqli_stmt_execute($stmt);
$res   = mysqli_stmt_get_result($stmt);
$count = mysqli_fetch_assoc($res)['cnt'];
mysqli_stmt_close($stmt);

if ($count >= $ev['capacity']) {
  $message  = 'Sorry, this event is full.';
  $msg_type = 'danger';
} else {
  $stmt = mysqli_prepare($conn, "SELECT registrationID FROM registrations WHERE studentID = ? AND eventID = ? AND status = 'cancelled'");
  mysqli_stmt_bind_param($stmt, 'si', $student_id, $event_id);
  mysqli_stmt_execute($stmt);
  $res       = mysqli_stmt_get_result($stmt);
  $cancelled = mysqli_fetch_assoc($res);
  mysqli_stmt_close($stmt);

  if ($cancelled) {
    $stmt = mysqli_prepare($conn, "UPDATE registrations SET status = 'confirmed', registrationDate = NOW() WHERE registrationID = ?");
    mysqli_stmt_bind_param($stmt, 'i', $cancelled['registrationID']);
  } else {
    $stmt = mysqli_prepare($conn, "INSERT INTO registrations (studentID, eventID, status) VALUES (?, ?, 'confirmed')");
    mysqli_stmt_bind_param($stmt, 'si', $student_id, $event_id);
  }

  if (mysqli_stmt_execute($stmt)) {
```

```
            $message  = 'You have successfully registered for this event!';
            $msg_type = 'success';
        } else {
            $message  = 'Registration failed. Please try again.';
            $msg_type = 'danger';
        }
        mysqli_stmt_close($stmt);
    }
  }
}
// get event details
$stmt = mysqli_prepare($conn, "SELECT e.*, o.name AS organizer_name, o.email AS organizer_email, o.role AS organizer_role FROM events e JOIN organizers o ON e.organizerID = o.organizerID WHERE e.eventID = ? AND (e.status = 'approved' OR EXISTS (SELECT 1 FROM approvals ap WHERE ap.eventID = e.eventID AND ap.decision = 'approved'))");
mysqli_stmt_bind_param($stmt, 'i', $event_id);
mysqli_stmt_execute($stmt);
$res   = mysqli_stmt_get_result($stmt);
$event = mysqli_fetch_assoc($res);
mysqli_stmt_close($stmt);

if (!$event) {
  header('Location: browse_events.php');
  exit;
}
// check if already registered
$stmt = mysqli_prepare($conn, "SELECT registrationID FROM registrations WHERE studentID = ? AND eventID = ? AND status != 'cancelled'");
mysqli_stmt_bind_param($stmt, 'si', $student_id, $event_id);
mysqli_stmt_execute($stmt);
mysqli_stmt_store_result($stmt);
$is_registered = mysqli_stmt_num_rows($stmt) > 0;
mysqli_stmt_close($stmt);
```

```
// get registration count
$stmt = mysqli_prepare($conn, "SELECT COUNT(*) AS cnt FROM registrations WHERE eventID = ? AND status = 'confirmed'");
mysqli_stmt_bind_param($stmt, 'i', $event_id);
mysqli_stmt_execute($stmt);
$res      = mysqli_stmt_get_result($stmt);
$reg_count = mysqli_fetch_assoc($res)['cnt'];
mysqli_stmt_close($stmt);

$spots_left = $event['capacity'] - $reg_count;
$is_full    = $spots_left <= 0;
$is_past    = strtotime($event['date']) < time();

$current_page = 'browse_events.php';
$student_name = $_SESSION['student_name'];
?>
```

11. PHP code for support

```
<?php
session_start();
require_once 'db.php';
$isStudent   = isset($_SESSION['student_id']);
$isOrganizer = isset($_SESSION['organizer_id']);
if (!$isStudent && !$isOrganizer) {
   header('Location: login.php');
   exit;
}
if ($isStudent) {
   $student_id = $_SESSION['student_id'];
   $stmt = mysqli_prepare($conn, "SELECT name FROM students WHERE studentID = ?");
   mysqli_stmt_bind_param($stmt, 's', $student_id);
   mysqli_stmt_execute($stmt);
   $row = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));
   mysqli_stmt_close($stmt);
```

```
    $userName     = $row['name'] ?? 'Student';
    $userSubtitle = $student_id;
    $_SESSION['student_name'] = $userName;
} else {
    $organizerID = (int)$_SESSION['organizer_id'];
    $row = mysqli_fetch_assoc(mysqli_query($conn,
        "SELECT name, role FROM organizers WHERE organizerID = $organizerID"));
    $userName      = $row['name'] ?? 'Organizer';
    $organizerRole = $row['role'] ?? '';
    $userSubtitle  = $organizerRole;
    $_SESSION['organizer_name'] = $userName;}
$success = '';
$errors  = [];
$categories = $isStudent
    ? ['Technical Issue','Registration Problem','Payment / Finance','Account & Access','Event Inquiry','Other']
    : ['Technical Issue','Event Approval Delay','Student Registration Problem','Payment / Finance','Account & Access','Venue / Logistics','Other'];
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $category = trim($_POST['category'] ?? '');
    $subject  = trim($_POST['subject']  ?? '');
    $message  = trim($_POST['message']  ?? '');
    if (!$category)           $errors[] = 'Please select a category.';
    if (!$subject)            $errors[] = 'Subject is required.';
    if (strlen($subject) > 255) $errors[] = 'Subject is too long (max 255 characters).';
    if (!$message)            $errors[] = 'Message is required.';
    if (strlen($message) < 20)  $errors[] = 'Message is too short (minimum 20 characters).';
    if (empty($errors)) {
        if ($isStudent) {
            $stmt = mysqli_prepare($conn,
                "INSERT INTO requests (studentID, category, subject, message, senderType, status)
                 VALUES (?, ?, ?, ?, 'student', 'Pending')");
            mysqli_stmt_bind_param($stmt, 'ssss', $student_id, $category, $subject, $message);
        } else {
            $stmt = mysqli_prepare($conn,
```

```
            "INSERT INTO requests (organizerID, category, subject, message, senderType, status)
             VALUES (?, ?, ?, ?, 'organizer', 'Pending')");
            mysqli_stmt_bind_param($stmt, 'isss', $organizerID, $category, $subject, $message);
        }
        if (mysqli_stmt_execute($stmt)) {
            $success = 'Your support request has been submitted. Admin will respond shortly.';
        } else {
            $errors[] = 'Database error: ' . mysqli_error($conn);
        }
        mysqli_stmt_close($stmt);
    }
}
if ($isStudent) {
    $sid_esc = mysqli_real_escape_string($conn, $student_id);
    $tickets = mysqli_query($conn, "
        SELECT r.*,
            (SELECT rp.reply FROM responses rp WHERE rp.requestID = r.requestID ORDER BY rp.replied_at DESC LIMIT 1) AS last_reply,
            (SELECT a.name  FROM responses rp JOIN admins a ON rp.adminID=a.adminID WHERE rp.requestID=r.requestID ORDER BY rp.replied_at DESC LIMIT 1) AS admin_name
        FROM requests r
        WHERE r.studentID = '$sid_esc' AND r.senderType = 'student'
        ORDER BY r.created_at DESC LIMIT 20");
} else {
    $tickets = mysqli_query($conn, "
        SELECT r.*,
            (SELECT rp.reply FROM responses rp WHERE rp.requestID = r.requestID ORDER BY rp.replied_at DESC LIMIT 1) AS last_reply,
            (SELECT a.name  FROM responses rp JOIN admins a ON rp.adminID=a.adminID WHERE rp.requestID=r.requestID ORDER BY rp.replied_at DESC LIMIT 1) AS admin_name
        FROM requests r
        WHERE r.organizerID = $organizerID AND r.senderType = 'organizer'
        ORDER BY r.created_at DESC LIMIT 20");
}$current_page = 'support.php';$page_title   = 'Support';?>
```

12. PHP for registrations page

```
<?php
session_start();
if (!isset($_SESSION['organizer_id'])) { header('Location: login.php'); exit; }
require_once 'db.php';
$organizerID  = (int)$_SESSION['organizer_id'];
$row = mysqli_fetch_assoc(mysqli_query($conn,
   "SELECT name, role FROM organizers WHERE organizerID = $organizerID"));
$organizerName = $row['name'] ?? 'Organizer';
$organizerRole = $row['role'] ?? '';
$_SESSION['organizer_name'] = $organizerName;
$filterEvent  = (int)($_GET['event']  ?? 0);
$search       = trim($_GET['search']  ?? '');
$filterStatus = $_GET['status']       ?? '';
$eventsQuery = mysqli_query($conn,
   "SELECT eventID, title FROM events WHERE organizerID = $organizerID ORDER BY date DESC");
$myEvents = [];
while ($ev = mysqli_fetch_assoc($eventsQuery)) $myEvents[] = $ev;

$where = "WHERE e.organizerID = $organizerID";
if ($filterEvent)  $where .= " AND r.eventID = $filterEvent";
if ($filterStatus) $where .= " AND r.status = '" . mysqli_real_escape_string($conn, $filterStatus) . "'";
if ($search) {
   $s = mysqli_real_escape_string($conn, $search);
   $where .= " AND (s.name LIKE '%$s%' OR s.email LIKE '%$s%' OR s.studentID LIKE '%$s%')";}
$registrations = mysqli_query($conn, "
   SELECT r.registrationID, r.registrationDate, r.status AS regStatus,
          s.studentID, s.name AS studentName, s.email, s.department AS studentDept,
          e.eventID, e.title AS eventTitle, e.date AS eventDate, e.capacity,
          e.is_paid, e.price,
          (SELECT COUNT(*) FROM registrations rx WHERE rx.eventID = e.eventID AND rx.status != 'cancelled') AS regCount
   FROM registrations r
```

```
    JOIN students s ON r.studentID = s.studentID
    JOIN events   e ON r.eventID   = e.eventID
    $where ORDER BY r.registrationDate DESC");
$totalReg    = (int)mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT COUNT(*) AS c FROM registrations r JOIN events e ON r.eventID=e.eventID
     WHERE e.organizerID=$organizerID AND r.status='confirmed'"))['c'];
$totalCancel = (int)mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT COUNT(*) AS c FROM registrations r JOIN events e ON r.eventID=e.eventID
     WHERE e.organizerID=$organizerID AND r.status='cancelled'"))['c'];

// CSV export — must run before any HTML
if (isset($_GET['export'])) {
    header('Content-Type: text/csv');
    header('Content-Disposition: attachment; filename="registrations_' . date('Ymd') . '.csv"');
    $out = fopen('php://output', 'w');
    fputcsv($out, ['Student ID','Name','Email','Department','Event','Event Date','Registered On','Fee','Status']);
    while ($r = mysqli_fetch_assoc($registrations)) {
        fputcsv($out, [
            $r['studentID'], $r['studentName'], $r['email'], $r['studentDept'],
            $r['eventTitle'], date('Y-m-d', strtotime($r['eventDate'])),
            date('Y-m-d H:i', strtotime($r['registrationDate'])),
            $r['is_paid'] === 'Yes' ? 'SAR ' . $r['price'] : 'Free',
            $r['regStatus']
        ]);
    }
    fclose($out);
    mysqli_close($conn);
    exit;
}

$current_page = 'registrations.php';
$page_title   = 'Registered Students';
?>
```

# 10. Test cases.

| SN | TEST NAME | STEPS | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| 1 | **Login (Valid User)** | 1. Open the website URL<br>2. Enter valid username or university ID<br>3. Enter correct password<br>4. Click Login | Redirect to dashboard based on role (student / organizer / admin) | Redirected to correct dashboard | **PASS** |
| 2 | **Student Registration** | 1. Click 'Create Account'<br>2. Fill name, username, email<br>3. Enter 9-digit university ID<br>4. Choose major and year<br>5. Set password (min 8 chars, 1 capital, 1 number)<br>6. Submit form | Account created, redirect to login page | Account created successfully | **PASS** |
| 3 | **Browse Events** | 1. Log in as student<br>2. Navigate to Browse Events<br>3. View list of approved events | Only approved events are shown with title, date, location, and capacity | Approved events listed correctly | **PASS** |
| 4 | **Register for Free Event** | 1. Log in as student<br>2. Open an approved free event<br>3. Click 'Register'<br>4. Confirm registration | Registration confirmed, event appears in My Registrations | Registration successful | **PASS** |

| | | | | | |
|---|---|---|---|---|---|
| 5 | **Pay & Register for Paid Event** | 1. Log in as student<br>2. Open an approved paid event<br>3. Click 'Pay & Register'<br>4. Enter valid card details (16-digit, CVV, expiry, name)<br>5. Submit payment | Payment accepted, registration confirmed, receipt shown | Payment and registration successful | **PASS** |
| 6 | **Payment (Invalid Card Details)** | 1. Navigate to a paid event<br>2. Click 'Pay & Register'<br>3. Enter card number with fewer than 16 digits<br>4. Submit | Error: 'Please fill in all card details correctly' | Validation error shown | **PASS** |
| 7 | **Cancel Registration** | 1. Log in as student<br>2. Go to My Registrations<br>3. Click Cancel on a confirmed registration | Registration status changed to Cancelled | Registration cancelled | **PASS** |
| 8 | **View My Registrations** | 1. Log in as student<br>2. Navigate to My Registrations | All past and current registrations shown with status (confirmed/cancelled) | Registrations listed correctly | **PASS** |
| 9 | **Add New Event** | 1. Log in as organizer<br>2. Click 'Add Event'<br>3. Fill title, description, date, location, capacity, department, gender, fee<br>4. Submit form | Event saved with status 'Pending', visible in My Events | Event created and pending approval | **PASS** |
| 10 | **Delete Event** | 1. Log in as organizer<br>2. Go to My Events<br>3. Click Delete on an event | Event and all its registrations removed | Event deleted successfully | **PASS** |

| | | | | | |
|---|---|---|---|---|---|
| | | 4. Confirm deletion prompt | | | |
| 11 | **View Registered Students** | 1. Log in as organizer<br>2. Go to Registered Students<br>3. Select an event from dropdown | List of confirmed and cancelled students shown with filters | Student list displayed | **PASS** |
| 12 | **Admin Dashboard Overview** | 1. Log in as admin<br>2. Navigate to Dashboard / Overview | Stats shown: total events, registrations, pending approvals, open requests | Dashboard stats displayed correctly | **PASS** |
| 13 | **Admin View Event Registrants** | 1. Log in as admin<br>2. Go to Events Management<br>3. Click 'Registrants' for any event | List of all registered students shown with confirmed/cancelled counts | Registrants list displayed | **PASS** |
| 14 | **Admin Manage Users** | 1. Log in as admin<br>2. Go to Users Management<br>3. View list of students and organizers | All users listed with name, ID, role | User list displayed correctly | **PASS** |

# 11. CONCLUSION & FUTURE WORK

## 11.1 CONCLUSION

The Development of University Event Management System (DUEMS) has been successfully built and delivered as a fully functional web-based platform designed to streamline event management at Jazan University. In this phase, the team moved from design into complete implementation, producing a working system that serves three distinct user roles: students, organizers, and administrators.

The implemented system includes a student portal for browsing and registering for events, an organizer dashboard for creating and managing events, and a comprehensive admin panel for approving events, managing users, overseeing registrations, and handling support requests. Core modules such as user authentication, event approval workflows, registration management, and support ticketing were all developed and integrated into a cohesive platform.

DUEMS represents a meaningful step toward the digital transformation of campus life at Jazan University. By replacing manual and fragmented processes with a centralized, role-based system, the platform improves efficiency, transparency, and the overall experience for everyone involved in university events. This project stands as a significant achievement in our academic journey and a practical contribution to the university community.

## 11.2 FUTURE WORK

While DUEMS delivers a solid and functional foundation, several enhancements are planned to extend its capabilities in future development cycles. One of the primary goals is to introduce full Arabic language support, allowing the platform to operate bilingually with a proper right-to-left layout. This enhancement will make the system significantly more accessible and inclusive for the Arabic-speaking members of the university community, ensuring that language is no longer a barrier to engaging with the platform.

Beyond language support, a dedicated payment gateway will be integrated to handle paid events, enabling students to complete secure online transactions directly through the platform. This module will include automated payment tracking and confirmation receipts, providing both students and organizers with a seamless, transparent, and trustworthy financial experience without the need for external tools or manual handling.

Furthermore, the system will be expanded to automatically generate and issue official attendance certificates upon event completion, as well as formal excuse forms that students can submit to their respective departments. This addition will bring meaningful academic value to student participation, recognizing attendance in a formal and institutionally recognized manner. Together, these planned enhancements reflect the team's commitment to continuous improvement and the long-term vision of delivering a platform that fully meets the evolving needs of Jazan University.

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